
Hierarchical Variable Selection for Nonconvex Penalized Cox Models in Individual-Participant-Data Meta-Analysis

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ABSTRACT

Individual-participant-data (IPD) meta-analysis of time-to-event outcomes must handle heterogeneity between studies both in baseline risk and in covariate effects, and increasingly must select covariates from many candidates. We propose a one-stage stratified Cox model in which each study keeps its own unspecified baseline hazard and each study-specific log-hazard ratio is the sum of a global effect and a sparse study-specific deviation, both unknown fixed parameters. MCP or SCAD penalties with separate tuning parameters act on the global effects and the deviations; a hierarchy constraint removes a covariate from every study when its global effect is zero, and a sharing rule identifies the decomposition. We give the objective function explicitly, a coordinate descent algorithm that respects the constraints exactly, and a two-dimensional cross-validation procedure with study-stratified folds, and we establish selection consistency and oracle efficiency for a dimension growing with the sample size. In simulations with 100 to 500 covariates, stratification and the nonconvex penalty drive the accuracy of global selection; the hierarchical deviations reduce the error of study-specific effects by a factor of 1.2 to 7.7 when heterogeneity is present, cost nothing when it is absent, and with the one-standard-error rule give the best global selection in most scenarios. In six ovarian cancer cohorts (1,446 patients, 500 genes), the method identifies two robust protective genes and shows that the association of a third is confined to one cohort. The method is implemented in the R package `hiermetacox`, and all code is publicly available. **Keywords:** Individual participant data meta-analysis | Stratified

Cox model | Between-study heterogeneity | Nonconvex penalty | Variable selection | Cross-validation | R package

1 Introduction

Individual participant data (IPD) meta-analysis pools the raw records of several studies instead of their published summaries (Riley et al., 2010, 2021). For time-to-event outcomes this allows a common definition of follow-up and censoring, adjustment for participant-level covariates, and

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analysis of covariate effects that published hazard ratios cannot provide (Tudur Smith et al., 2005; Debray et al., 2015; Krishan and Dwan, 2025). Two strategies are used. A two-stage analysis fits a Cox model within each study and combines the estimated log-hazard ratios by fixed- or random-effects meta-analysis (DerSimonian and Laird, 1986); a one-stage analysis fits a single model to all participants while respecting their clustering within studies (Bowden et al., 2011; Burke et al., 2017). The two agree when studies are large, but the one-stage approach is more reliable when some studies contribute few events, and it is the natural setting for joint modelling of many covariates (Burke et al., 2017).

Handling between-study heterogeneity in one-stage survival models. Studies differ in recruitment, treatment and follow-up, so both the baseline hazard and the covariate effects may vary. Heterogeneity in the baseline hazard is usually handled either by *stratification*, which gives each study its own unspecified baseline hazard in the partial likelihood (Therneau and Grambsch, 2000; Crowther et al., 2012), or by a *shared frailty*, a random study effect acting multiplicatively on a common baseline hazard (Hougaard, 2000; Duchateau and Janssen, 2008). Heterogeneity in covariate effects is handled by random coefficients (random slopes) in mixed-effects Cox models (Vaida and Xu, 2000; Ripatti and Palmgren, 2000; Therneau et al., 2003) or in parametric multilevel survival models (Crowther et al., 2014); these are the one-stage counterparts of the random-effects meta-analysis model of DerSimonian and Laird (1986). In IPD meta-analysis of time-to-event outcomes such models have been used to estimate the between-trial variance of a treatment effect and of treatment–covariate interactions (Tudur Smith et al., 2005; Bowden et al., 2011). Estimation typically uses penalised partial likelihood, the hierarchical likelihood (Ha et al., 2001), or Bayesian computation (Legrand et al., 2005), and is implemented, for example, in `coxme` and `frailtypack` (Rondeau et al., 2012). All of these methods estimate a small, pre-specified set of effects; they do not select covariates.

Variable selection for clustered and multi-study survival data. Penalised partial likelihood is the standard tool for selecting covariates in the Cox model: the lasso (Tibshirani, 1997), the adaptive lasso (Zhang and Lu, 2007), and the nonconvex SCAD (Fan and Li, 2001) and MCP (Zhang, 2010) penalties, whose oracle properties for the Cox model were established by Fan and Li (2002) for fixed dimension and by Bradic et al. (2011) and Huang et al. (2013) in high dimensions; coordinate descent makes them routine (Simon et al., 2011; Breheny and Huang, 2011). Fan and Li (2002) also treated the frailty model, and selection in frailty models has since been studied with penalised likelihood (Androulakis et al., 2012), the penalised h -likelihood (Ha et al., 2014), and group lasso penalties (Utazirubanda et al., 2021). Groll et al. (2017) went further and selected random slopes, so that a covariate effect can be declared study-specific or common. These frailty approaches assume a parametric (usually Gaussian or gamma) distribution for the study effects, which is hard to check with the five to fifteen studies typical of an IPD meta-analysis, and they are computationally demanding when many covariates are candidates.

A second line of work treats the study-specific coefficients as fixed parameters. Ma et al. (2011) and He et al. (2016) proposed group penalties that select a covariate jointly in all studies; Cheng et al. (2015) decomposed study-specific Cox coefficients into a common part and a study-specific part and used adaptive lasso penalties to separate homogeneous from heterogeneous covariates in pooled cohort studies; Gross and Tibshirani (2016) and Ollier and Viallon (2017) studied the same decomposition for linear and generalised linear models with the lasso, and Tang and Song

(2016) fused coefficients across studies. None of these combines (i) a stratified partial likelihood, (ii) nonconvex penalties with separate tuning of common and study-specific effects, and (iii) an explicit rule that a covariate removed from the common model is removed from every study.

Contribution. We propose a one-stage stratified Cox model in which each study-specific log-hazard ratio is written as $\theta_{kj} = \alpha_j + \varepsilon_{kj}$, a shared global effect plus a sparse study-specific deviation, both treated as unknown fixed parameters. SCAD or MCP penalties with two tuning parameters are placed on the global effects and on the deviations, and a hierarchy constraint enforces that $\alpha_j = 0$ implies $\varepsilon_{kj} = 0$ in every study. Our contributions are:

- (i) an explicit objective function and identification rule for the decomposition, and its relation to frailty models (Section 2);
- (ii) a coordinate descent algorithm that respects the hierarchy exactly, together with a fully specified two-dimensional cross-validation procedure for the two tuning parameters (Section 3);
- (iii) selection consistency and oracle efficiency when the dimension grows with the sample size (Section 4);
- (iv) a simulation study with p between 100 and 500, including settings in which p exceeds every study's sample size, that separates the contributions of stratification, the penalty and the hierarchy (Section 5);
- (v) an analysis of six ovarian cancer cohorts with 500 candidate genes that reports hazard ratios with confidence intervals, selection stability, sensitivity to a frailty specification, and cross-study validation (Section 6);
- (vi) the open-source R package `hiermetacox` and the complete code reproducing every table and figure (Section 3.4).

2 Model and estimator

2.1 Data and notation

We observe IPD from K independent studies. Study k ($k = 1, \dots, K$) has n_k participants and $N = \sum_k n_k$. For participant i of study k let $\mathbf{x}_{ki} \in \mathbb{R}^p$ be the covariate vector, T_{ki} the event time and C_{ki} the censoring time. We observe $Y_{ki} = \min(T_{ki}, C_{ki})$ and $\delta_{ki} = \mathbf{1}(T_{ki} \leq C_{ki})$. Censoring is assumed non-informative within each study: T_{ki} and C_{ki} are independent given $\mathbf{x}_{ki}$ (Kalbfleisch and Prentice, 2002). The risk set of study k at time t is $\mathcal{R}_k(t) = \{l : Y_{kl} \geq t\}$; it contains participants of study k only. Covariates are standardised within each study to mean zero and unit variance, so every coefficient below is a log-hazard ratio per within-study standard deviation.

2.2 Hierarchical decomposition

The hazard of participant i in study k is

$$h_{ki}(t \mid \mathbf{x}_{ki}) = h_{0k}(t) \exp(\mathbf{x}_{ki}^\top \boldsymbol{\theta}_k), \quad \boldsymbol{\theta}_k = \boldsymbol{\alpha} + \boldsymbol{\varepsilon}_k, \quad (2.1)$$

that is, $\theta_{kj} = \alpha_j + \varepsilon_{kj}$ for $j = 1, \dots, p$. The components of (2.1) are as follows.

- $h_{0k}(\cdot)$ is the baseline hazard of study k . It is an *unknown, unspecified function*, different for every study and not estimated; it is eliminated by the stratified partial likelihood.
- $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_p)^\top$ is the vector of *global effects*. It is an *unknown, fixed* (non-random)

parameter to be estimated. α_j is the log-hazard ratio of covariate j shared by the studies that do not deviate from it.

- $\boldsymbol{\varepsilon}_k = (\varepsilon_{k1}, \dots, \varepsilon_{kp})^\top$ is the vector of *study-specific deviations* of study k . The $\boldsymbol{\varepsilon}_k$ are also *unknown, fixed* parameters; they are not random effects and no distribution is assumed for them. $\varepsilon_{kj} = 0$ means that study k shares the global effect of covariate j .

The decomposition is not identified without a rule, because $(\alpha_j + c, \varepsilon_{1j} - c, \dots, \varepsilon_{Kj} - c)$ gives the same θ_{kj} for every c . The original version of this work imposed the zero-sum constraint $\sum_k \varepsilon_{kj} = 0$; we abandon it here because it is incompatible with sparse deviations (a single deviating study would force all others to deviate as well). We use instead the *sharing rule*:

$$\text{for each } j \text{ with } \alpha_j \neq 0, \quad |\{k : \varepsilon_{kj} = 0\}| \geq 1, \quad (2.2)$$

and, among the decompositions satisfying (2.2), the one with the smallest penalty (Section 2.4) is chosen. When a strict majority of studies shares a common value of θ_{kj} , this rule returns that common value as α_j , so that α_j is the effect in the “typical” study and the deviations single out the discordant ones.

The second structural assumption is the *hierarchy*: a covariate that has no global effect has no study-specific effect either,

$$\alpha_j = 0 \implies \varepsilon_{kj} = 0 \quad \text{for all } k. \quad (2.3)$$

We write $\mathcal{H}$ for the set of $(\boldsymbol{\alpha}, \boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_K)$ satisfying (2.2) and (2.3). Under (2.3) a covariate is either absent from the model or present in every study with effect α_j , possibly modified in some studies. A covariate that acts in only a minority of studies is therefore represented by a small α_j with large deviations; the fitted deviations make such a pattern visible (Section 6).

When $\boldsymbol{\varepsilon}_k = \mathbf{0}$ for all k , model (2.1) is the usual stratified Cox model with common coefficients. When the $\boldsymbol{\varepsilon}_k$ are unrestricted, it is equivalent to K separate Cox models. The decomposition interpolates between these two extremes, and the penalty decides, covariate by covariate and study by study, where on this range the data lie.

2.3 Relation to frailty and random-effects models

Model (2.1) handles the two kinds of heterogeneity differently from a frailty model. Baseline heterogeneity is absorbed by stratification rather than by a random intercept $h_{0k}(t) = u_k h_0(t)$; stratification does not require the baseline hazards to be proportional across studies and needs no distributional assumption for u_k (Therneau and Grambsch, 2000). Effect heterogeneity is represented by fixed deviations ε_{kj} rather than by random slopes $b_{kj} \sim N(0, \tau_j^2)$ as in Vaida and Xu (2000), Ripatti and Palmgren (2000) and Groll et al. (2017). The random-slope model shrinks every study towards the mean by an amount governed by τ_j^2 , estimated from K values; our penalty instead sets most deviations exactly to zero and leaves large ones essentially unshrunk, which is the behaviour sought when the question is *which* studies differ. Fixed deviations also avoid an integral over the random effects, so the estimator remains a penalised partial likelihood and scales to hundreds of covariates. The price is that inference is conditional on the studies at hand; prediction for a new study uses $\boldsymbol{\alpha}$ only. In Section 6 we compare hazard ratios from the stratified model with those from a shared gamma frailty model.

2.4 Objective function

The stratified log partial likelihood of study k , with Breslow's treatment of ties, is

$$\ell_k(\boldsymbol{\theta}_k) = \sum_{i=1}^{n_k} \delta_{ki} \left[\mathbf{x}_{ki}^\top \boldsymbol{\theta}_k - \log \sum_{l \in \mathcal{R}_k(Y_{ki})} \exp(\mathbf{x}_{kl}^\top \boldsymbol{\theta}_k) \right]. \quad (2.4)$$

The estimator is

$$(\hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\varepsilon}}_1, \dots, \hat{\boldsymbol{\varepsilon}}_K) = \arg \min_{(\boldsymbol{\alpha}, \boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_K) \in \mathcal{H}} \mathcal{Q}_\lambda(\boldsymbol{\alpha}, \boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_K), \quad (2.5)$$

with objective function

$$\begin{aligned} \mathcal{Q}_\lambda = & -\frac{1}{N} \sum_{k=1}^K \ell_k(\boldsymbol{\alpha} + \boldsymbol{\varepsilon}_k) + \sum_{j=1}^p P(|\alpha_j|; \lambda_\alpha, \gamma) \\ & + \sum_{k=1}^K \sum_{j=1}^p P(\omega_k |\varepsilon_{kj}|; \lambda_\varepsilon, \gamma), \end{aligned} \quad (2.6)$$

where $\omega_k = (n_k/N)^{1/2}$. The three terms are: the negative stratified partial log-likelihood scaled by N (each study keeps its own risk sets and baseline hazard); a penalty on each global effect with tuning parameter $\lambda_\alpha > 0$; and a penalty on each deviation with tuning parameter $\lambda_\varepsilon > 0$. The weight ω_k is the standard deviation of the column $\mathbf{x}_{\cdot j} \mathbf{1}(\text{study} = k)$ of the design matrix that carries ε_{kj} , so that penalising $\omega_k |\varepsilon_{kj}|$ is equivalent to the usual practice of penalising coefficients of standardised columns; it gives global effects and deviations comparable curvature in the coordinate-wise problems of Section 3 and is the weighting advocated for the same decomposition by [Gross and Tibshirani \(2016\)](#). The penalty P is SCAD ([Fan and Li, 2001](#)),

$$P(t; \lambda, \gamma) = \begin{cases} \lambda t, & t \leq \lambda, \\ \frac{2\gamma\lambda t - t^2 - \lambda^2}{2(\gamma - 1)}, & \lambda < t \leq \gamma\lambda, \\ \lambda^2(\gamma + 1)/2, & t > \gamma\lambda, \end{cases}$$

with $\gamma = 3.7$, or MCP ([Zhang, 2010](#)), $P(t; \lambda, \gamma) = \lambda t - t^2/(2\gamma)$ for $t \leq \gamma\lambda$ and $\gamma\lambda^2/2$ otherwise, with $\gamma = 3$. Both equal the lasso penalty λt near zero, which produces exact zeros, and are constant beyond $\gamma\lambda$, so large effects are not shrunk. Replacing P by λt gives a convex hierarchical lasso, which we include as a comparator.

Because $P(\cdot; \lambda, \gamma)$ is nonconvex, (2.5) may have several local minimisers; as is standard for folded concave penalties ([Fan and Li, 2001](#); [Zhang, 2010](#); [Fan et al., 2014](#)), the estimator is the local minimiser reached by the pathwise algorithm of Section 3, started from the null model at the largest λ_α and warm-started along the path.

3 Computation and tuning

3.1 Algorithm

Write $\eta_{ki} = \mathbf{x}_{ki}^\top (\boldsymbol{\alpha} + \boldsymbol{\varepsilon}_k)$ for the linear predictor and $L(\boldsymbol{\eta}) = -N^{-1} \sum_k \ell_k$. At the current iterate $\tilde{\boldsymbol{\eta}}$, L is replaced by the quadratic surrogate

$$L(\boldsymbol{\eta}) \approx L(\tilde{\boldsymbol{\eta}}) + \frac{1}{2} \sum_{k,i} w_{ki} (z_{ki} - \eta_{ki})^2, \quad (3.1)$$

with weights $w_{ki} = N^{-1} e^{\tilde{\eta}_{ki}} A_{ki}$ and working responses $z_{ki} = \tilde{\eta}_{ki} + (\delta_{ki} - e^{\tilde{\eta}_{ki}} A_{ki}) / (e^{\tilde{\eta}_{ki}} A_{ki})$, where $A_{ki} = \sum_{i': Y_{ki'} \leq Y_{ki}} \delta_{ki'} / \sum_{l \in \mathcal{R}_k(Y_{ki'})} e^{\tilde{\eta}_{kl}}$ is the Breslow cumulative hazard increment *within study* k . This is the diagonal upper bound on the Hessian of the partial likelihood used by Simon et al. (2011) and Breheny and Huang (2011); stratification enters only through the fact that all risk-set sums run over one study. All quantities are computed in $O(N)$ operations by cumulative sums after sorting by study and time.

The penalised surrogate is minimised by cycling over covariates. For covariate j the block $(\alpha_j, \varepsilon_{1j}, \dots, \varepsilon_{Kj})$ is updated as follows.

- (a) *Global effect.* Given the deviations, the surrogate is a one-dimensional function $\frac{v}{2}(\alpha_j - z)^2 + P(|\alpha_j|; \lambda_\alpha, \gamma)$, with $v = \sum_{k,i} w_{ki} x_{kij}^2$ and z the unpenalised coordinate minimiser. Its *global* minimiser is computed exactly by comparing the stationary points of each branch of P with the branch boundaries and zero; this remains valid when $v \leq 1/\gamma$, where the usual closed-form thresholding rule does not apply.
- (b) *Hierarchy.* If the update in (a) gives $\alpha_j = 0$ while some $\varepsilon_{kj} \neq 0$, the pair is infeasible. The algorithm then compares the current block with the all-zero block and keeps whichever has the smaller surrogate objective. Every iterate therefore satisfies (2.3).
- (c) *Deviations.* If $\alpha_j \neq 0$, each ε_{kj} is updated by the same exact one-dimensional rule using study k only, with penalty $P(\omega_k |\cdot|; \lambda_\varepsilon, \gamma)$.
- (d) *Re-centring.* The move $(\alpha_j + c, \varepsilon_{\cdot j} - c)$ leaves every θ_{kj} , and hence the likelihood, unchanged. Its penalty is concave (SCAD, MCP) or linear (lasso) in c between the breakpoints $c \in \{\varepsilon_{1j}, \dots, \varepsilon_{Kj}\}$, so the minimum over c is attained at one of them. The algorithm evaluates the breakpoints and moves to the best one, which enforces the sharing rule (2.2).

Steps (a)–(d) never increase the penalised surrogate. Because the surrogate is not a global majoriser of L , the new iterate is accepted only if the true objective (2.6) has not increased; otherwise the step is halved until it has. After a full sweep, sweeps are restricted to covariates with $\alpha_j \neq 0$ until convergence, followed by a full sweep to check for new entries. Algorithm 1 summarises the procedure.

3.2 Choice of the two tuning parameters

The pair $(\lambda_\alpha, \lambda_\varepsilon)$ is chosen jointly by V -fold cross-validation over a two-dimensional grid. We parametrise the grid by λ_α and the ratio $r = \lambda_\varepsilon / \lambda_\alpha$, because r has a direct meaning: small r makes deviations cheap relative to global effects, large r pushes the model towards common coefficients. The procedure is:

1. *Grid.* $M = 20$ values of λ_α , log-spaced from $\lambda_{\max}$ to $\kappa \lambda_{\max}$ with $\kappa = 0.05$, and $r \in \{0.5, 1, 2\}$; the path is truncated once more than $d_{\max} = 100$ global effects are active.
2. *Folds.* Participants are assigned to V folds separately within each study and, within study, separately for events and censored observations. Every training set therefore contains every study, which is necessary because $\boldsymbol{\varepsilon}_k$ is estimated from study k only, and every fold has a similar event

Algorithm 1 Pathwise fit of (2.5) for a fixed ratio $r = \lambda_\varepsilon/\lambda_\alpha$

```

1: Standardise covariates within study; sort by study and time.
2:  $\lambda_{\max} \leftarrow \max_j |\partial L(\mathbf{0})/\partial \alpha_j|$ ; grid  $\lambda_{\max} = \lambda^{(1)} > \dots > \lambda^{(M)} = \kappa \lambda_{\max}$ , log-spaced.
3:  $(\alpha, \varepsilon) \leftarrow (\mathbf{0}, \mathbf{0})$ 
4: for  $m = 1, \dots, M$  do ▷ warm start from  $m - 1$ 
5:    $\lambda_\alpha \leftarrow \lambda^{(m)}$ ,  $\lambda_\varepsilon \leftarrow r \lambda^{(m)}$ 
6:   repeat ▷ outer iteration
7:     Compute  $w_{ki}, z_{ki}$  at the current  $\eta$  (study-wise risk sets).
8:     repeat
9:       Sweep  $j = 1, \dots, p$  (full sweep, then active set only),
10:      applying the block update (a)–(d).
11:    until max. change in any coefficient  $< \tau_{\text{in}}$ 
12:    If  $\mathcal{Q}_\lambda$  increased, halve the step until it does not.
13:  until max. change in any coefficient  $< \tau_{\text{out}}$ 
14:  Store  $(\hat{\alpha}^{(m)}, \hat{\varepsilon}^{(m)})$ ; stop if  $|\{j : \hat{\alpha}_j \neq 0\}| > d_{\max}$ .
15: end for

```

proportion.

3. *Criterion.* For every grid point (λ_α, r) and fold v , the model is fitted on the other folds and the linear predictors $\hat{\eta}_{ki}^{(-v)} = \mathbf{x}_{ki}^\top \hat{\boldsymbol{\theta}}_k^{(-v)}$ of the held-out participants are computed. After all folds, every participant has an out-of-fold linear predictor and the criterion is the cross-validated stratified deviance

$$\text{CV}(\lambda_\alpha, r) = -2 \sum_{k=1}^K \ell_k(\hat{\boldsymbol{\eta}}_k^{\text{cv}}), \quad (3.2)$$

the partial likelihood (2.4) of the full data, evaluated at the out-of-fold linear predictors, with study-specific risk sets. This is the linear-predictor approach of Dai and Breheny (2019), which is more stable than the fold-by-fold approach of Verweij and van Houwelingen (1993) when folds contain few events.

4. *Selection.* The *minimum rule* takes the grid point minimising (3.2). The *one-standard-error rule* computes the deviance of each fold on its own risk sets, takes $\text{SE} = V^{1/2}$ times their standard deviation at the minimum, and chooses, among grid points with $\text{CV} \leq \text{CV}_{\min} + \text{SE}$, the model with the fewest nonzero parameters (global effects plus deviations), breaking ties by larger λ_α and then larger r .

Recommendations. We recommend $V = 5$ folds, which leaves about 80% of every study in each training set and is what we use throughout; with small studies (fewer than about 20 events in the smallest study) $V = 10$ keeps training sets closer to the full data at twice the cost, and we report the sensitivity of the selected genes to this choice in Section 6. We recommend the one-standard-error rule when the aim is to select variables and the minimum rule when the aim is to estimate study-specific effects or predict; the simulation in Section 5 supports both recommendations. Both rules are read off the same cross-validation run.

3.3 Inference after selection

For the selected support $\hat{\mathcal{S}} = \{\alpha_j : \hat{\alpha}_j \neq 0\} \cup \{\varepsilon_{kj} : \hat{\varepsilon}_{kj} \neq 0\}$ we refit the unpenalised stratified Cox model with one column per element of $\hat{\mathcal{S}}$ and report Wald standard errors, hazard ratios $\exp(\hat{\alpha}_j)$ with 95% confidence intervals and p -values, together with study-specific hazard ratios $\exp(\hat{\theta}_{kj})$. By Theorems 4.1 and 4.2, these intervals are asymptotically valid on the event of correct selection, and

the sharing rule guarantees that the refit design has full column rank. In finite samples they are conditional on the selected model and ignore selection uncertainty (Berk et al., 2013); we therefore also report the proportion of $B = 100$ within-study bootstrap resamples, refitted at the selected tuning parameters, in which each covariate is selected (Efron and Tibshirani, 1993; Meinshausen and Bühlmann, 2010), and we evaluate the coverage of the Wald intervals by simulation in Section 5.

3.4 Software and cost

The method is implemented in the R package `hiermetacox`, available with all code for this paper at <https://github.com/EmmanuelMasavoDjegou/Hierarchical-Variable-Selection-in-Penalized-Cox-Models-for-IPD-Meta-Analysis>. The solver is written in C. The main functions are `hmcox()` (path fit), `cv.hmcox()` (two-dimensional cross-validation with both selection rules), `hmcox_inference()` (post-selection Wald inference), `hmcox_boot()` (bootstrap selection frequencies) and `sim_ipd()` (the simulation design of Section 5). A typical call is

```
fit <- cv.hmcox(X, time, status, study, penalty = "MCP",
               nfolds = 5, ratio = c(0.5, 1, 2), rule = "lse")
coef(fit); hmcox_inference(fit)
```

The package tests check that the unpenalised solver reproduces `survival::coxph` with `strata` to numerical precision, that the lasso path matches the stratified Cox model of `glmnet`, that SCAD and MCP solutions satisfy the Karush–Kuhn–Tucker conditions, and that every fitted model lies in $\mathcal{H}$. One sweep costs $O(Np)$ operations; the cost of cross-validation is $(V + 1) \times |\{r\}|$ path fits. In the simulation, five-fold cross-validation over the full two-dimensional grid took 3 to 8 seconds per data set for hierarchical MCP on one core (up to $N = 1,000$ and $p = 500$), against 0.3 to 3 seconds for the one-dimensional common-effect fit; for the application ($N = 1,446$, $p = 500$, $K = 6$) it took 18.5 seconds (Table 5).

4 Theoretical properties

The number of studies K is fixed and $N \rightarrow \infty$ with $n_k/N \rightarrow \pi_k \in (0, 1)$, so that $\omega_k \rightarrow \pi_k^{1/2}$. The number of covariates $p = p_N$ may grow with N . Collect the parameters in $\boldsymbol{\vartheta} = (\boldsymbol{\alpha}^\top, \boldsymbol{\varepsilon}_1^\top, \dots, \boldsymbol{\varepsilon}_K^\top)^\top \in \mathbb{R}^{(K+1)p}$ and let $\boldsymbol{\vartheta}_0$ be the true value, defined by the sharing rule of Section 2.2. Let $\mathcal{A}_0 = \{j : \alpha_{0j} \neq 0\}$ with $s_\alpha = |\mathcal{A}_0|$, $\mathcal{A}_0^{(k)} = \{j : \varepsilon_{0,kj} \neq 0\}$ with $s_k = |\mathcal{A}_0^{(k)}|$, $\mathcal{S}_0$ the set of indices of $\boldsymbol{\vartheta}$ that are nonzero at $\boldsymbol{\vartheta}_0$, $s_0 = |\mathcal{S}_0|$ and $d_N = \max(s_\alpha, \max_k s_k)$.

Assumption 1 (Structure). (i) *Hierarchy*: $\mathcal{A}_0^{(k)} \subseteq \mathcal{A}_0$ for every k . (ii) *Identification*: for every $j \in \mathcal{A}_0$ the set $\{k : \varepsilon_{0,kj} = 0\}$ contains more than $K/2$ studies.

Assumption 2 (Censoring). For each k , $T_{ki} \perp C_{ki} \mid \mathbf{x}_{ki}$, and with τ_k the end of follow-up there is $c_0 > 0$ with $\Pr(Y_{ki} \geq \tau_k \mid \mathbf{x}_{ki}) \geq c_0$.

Assumption 3 (Baseline hazards). There is $\Lambda < \infty$ with $0 \leq h_{0k}(t) \leq \Lambda$ for $t \in [0, \tau_k]$ and all k .

Assumption 4 (Covariates). There is $M < \infty$ with $\|\mathbf{x}_{ki}\|_\infty \leq M$ almost surely for all k, i, N .

Assumption 5 (Information). Let $\mathbf{I}_{N,\mathcal{S}}(\boldsymbol{\vartheta})$ be the expected information of the stratified partial likelihood with respect to the coordinates of $\boldsymbol{\vartheta}$ in $\mathcal{S}$. There are $0 < c_1 < c_2 < \infty$ with $c_1 \leq \lambda_{\min}(N^{-1}\mathbf{I}_{N,\mathcal{S}_0}(\boldsymbol{\vartheta}_0)) \leq \lambda_{\max}(N^{-1}\mathbf{I}_{N,\mathcal{S}_0}(\boldsymbol{\vartheta}_0)) \leq c_2$ for all large N , and the score and Hessian admit the usual uniform expansions in a neighbourhood of $\boldsymbol{\vartheta}_0$ (Andersen and Gill, 1982).

Assumption 6 (Penalty). $P(\cdot; \lambda, \gamma)$ is SCAD or MCP with fixed γ ; hence $P'(0+; \lambda, \gamma) = \lambda$ and $P'(t; \lambda, \gamma) = 0$ for $t > \gamma\lambda$.

Assumption 7 (Rates). As $N \rightarrow \infty$, $\lambda_\alpha, \lambda_\varepsilon \rightarrow 0$,

$$\sqrt{N/\log p_N} \min(\lambda_\alpha, \lambda_\varepsilon) \rightarrow \infty,$$

$$\min_{j \in \mathcal{A}_0} |\alpha_{0j}| \gg \max(\lambda_\alpha, d_N^{1/2} N^{-1/2}),$$

$$\min_{k, j \in \mathcal{A}_0^{(k)}} |\varepsilon_{0,kj}| \gg \max(\lambda_\varepsilon, d_N^{1/2} N^{-1/2}).$$

Assumption 8 (Dimension). $p_N = O(N^c)$ for some $c < 1$ and $d_N = o(N^{1/2})$.

Assumption 1(ii) is what makes the columns of the design restricted to $\mathcal{S}_0$ linearly independent: for every active j at least one study has no deviation column, so the global column $\mathbf{x}_{\cdot j}$ is not the sum of the deviation columns. Without it the information in Assumption 5 would be singular. The majority requirement also makes $\boldsymbol{\vartheta}_0$ the unique sparsest decomposition of $(\boldsymbol{\theta}_{01}, \dots, \boldsymbol{\theta}_{0K})$. Assumption 7 is compatible with Assumption 8: since $\log p_N = O(\log N)$, tuning parameters with $\sqrt{\log p_N/N} \ll \lambda \ll 1$ exist, and minimum signals may decrease to zero slowly. These conditions follow Fan and Li (2002) and Bradic et al. (2011); the growing-dimension arguments follow He and Shao (2000) and Wang (2011).

Theorem 4.1 (Selection consistency). Under Assumptions 1–8, with probability tending to one there is a local minimiser $\hat{\boldsymbol{\vartheta}} \in \mathcal{H}$ of (2.6) such that $\Pr(\hat{\mathcal{A}} = \mathcal{A}_0) \rightarrow 1$ and, for each k , $\Pr(\hat{\mathcal{A}}^{(k)} = \mathcal{A}_0^{(k)}) \rightarrow 1$, where $\hat{\mathcal{A}} = \{j : \hat{\alpha}_j \neq 0\}$ and $\hat{\mathcal{A}}^{(k)} = \{j : \hat{\varepsilon}_{kj} \neq 0\}$.

Theorem 4.2 (Oracle efficiency). Under the assumptions of Theorem 4.1, on the event of correct selection, $\sqrt{N} \mathbf{I}_{\mathcal{S}_0}^{1/2}(\boldsymbol{\vartheta}_0)(\hat{\boldsymbol{\vartheta}}_{\mathcal{S}_0} - \boldsymbol{\vartheta}_{0,\mathcal{S}_0}) \rightarrow \mathcal{N}(\mathbf{0}, \mathbf{I}_{\mathcal{S}_0})$ in distribution when $\mathcal{S}_0$ is fixed, where $\mathbf{I}_{\mathcal{S}_0}(\boldsymbol{\vartheta}_0) = \lim N^{-1} \mathbf{I}_{N,\mathcal{S}_0}(\boldsymbol{\vartheta}_0)$. In particular $\hat{\boldsymbol{\alpha}}_{\mathcal{A}_0}$ and each $\hat{\boldsymbol{\varepsilon}}_{k,\mathcal{A}_0^{(k)}}$ are asymptotically as efficient as the unpenalised stratified Cox estimator that knows $\mathcal{S}_0$.

Corollary 4.1 (Hierarchical coherence). Under the assumptions of Theorem 4.1, $\Pr(\hat{\mathcal{A}}^{(k)} \subseteq \hat{\mathcal{A}} \text{ for all } k) \rightarrow 1$. Moreover every iterate of Algorithm 1 lies in $\mathcal{H}$, so for the computed estimator the inclusion holds with probability one.

Proposition 4.1 (Algorithm). The iterates of Algorithm 1 for fixed $(\lambda_\alpha, \lambda_\varepsilon)$ satisfy (i) $\boldsymbol{\vartheta}^{(t)} \in \mathcal{H}$ for all t ; (ii) $\mathcal{Q}_\lambda(\boldsymbol{\vartheta}^{(t+1)}) \leq \mathcal{Q}_\lambda(\boldsymbol{\vartheta}^{(t)})$; (iii) $\mathcal{Q}_\lambda(\boldsymbol{\vartheta}^{(t)})$ converges. If the iterates converge, the limit $\bar{\boldsymbol{\vartheta}}$ is a fixed point of the block update at its own surrogate, and hence satisfies the first-order conditions of $\mathcal{Q}_\lambda$ for every coordinate whose hierarchy constraint is not active.

Theorem 4.2 justifies the Wald intervals of Section 3.3 asymptotically; Proposition 4.1 is deliberately limited to what the algorithm guarantees. It does not assert that the computed local solution is the one of Theorem 4.1; for folded concave penalties this can be established for specific initialisations (Fan et al., 2014), and we assess the computed estimator by simulation. The hierarchy constraint is active when α_j would be zero but some deviations would not; then α_j is kept at a small nonzero value and the first-order condition for α_j holds only as an inequality. Extension to $p \gg N$ would require a screening step such as sure independence screening (Fan and Lv, 2008; Fan et al., 2010).

5 Simulation study

5.1 Design

Data are generated from model (2.1) by `sim_ipd()`. In the base configuration there are $K = 5$ studies of $n_k = 200$ participants ($N = 1,000$) and $p = 100$ covariates, drawn from a multivariate normal distribution with AR(1) correlation $\rho^{|j-l|}$, $\rho = 0.5$. Six covariates have global effects $\boldsymbol{\alpha}_{\mathcal{A}_0} = (1, -0.8, 0.6, -0.5, 0.4, -0.3)$ and the other 94 have none. Heterogeneity is sparse: for each of the first four covariates, two of the five studies, chosen at random, receive a deviation $\varepsilon_{kj} \sim N(0, \sigma_\varepsilon^2)$ with $\sigma_\varepsilon = 0.4$, and all other deviations are zero;

the three remaining studies share α_j , so the sharing rule (2.2) identifies the truth. Study k has a Weibull baseline hazard with shape 1.5 and scale drawn from $\text{Uniform}(0.05, 0.15)$, and event times are generated by inversion (Bender et al., 2005). Censoring times are exponential and independent of the covariates; their rate is chosen separately in each study by root-finding so that the expected censoring proportion equals the target of 30% (Wan, 2017).

Twelve further scenarios change one feature of the base configuration at a time (Table 1). They vary the correlation, the size and extent of heterogeneity, censoring, signal strength, the dimension up to $p = 500$ (so that p exceeds every n_k and equals half of N), the number of studies, and the balance of study sizes. The *dense heterogeneity* scenario, in which all five studies deviate, violates the identification assumption and is included to show what happens when the model is misspecified in this respect. Each scenario is replicated $R = 25$ times; replicate r of scenario s uses the seed $20260000 + 1000s + r$.

Table 1: Simulation scenarios. The base configuration has $K = 5$, $n_k = 200$, $p = 100$, $\rho = 0.5$, $\alpha_{\mathcal{A}_0} = (1, -0.8, 0.6, -0.5, 0.4, -0.3)$, deviations on covariates 1–4 in two of five studies with $\sigma_\varepsilon = 0.4$, and 30% censoring.

#	Scenario	Change from the base configuration
1	Base	Reference configuration
2	Independent covariates	$\rho = 0$
3	Strong correlation	$\rho = 0.8$
4	No heterogeneity	$\sigma_\varepsilon = 0$ (no deviations)
5	Strong heterogeneity	$\sigma_\varepsilon = 0.8$
6	Dense heterogeneity	all K studies deviate on covariates 1–4
7	Heavy censoring	censoring 60%
8	Weak signals	$\alpha_{\mathcal{A}_0}$ halved
9	$p = 200$	$p = 200$
10	$p = 500$	$p = 500$
11	Ten studies	$K = 10$, $n_k = 100$
12	Unequal study sizes	$n_k = 100, 150, 200, 250, 300$
13	Small studies, $p > n_k$	$n_k = 100$, $p = 200$ ($p > n_k$)

5.2 Methods compared

We compare eight estimators, chosen to isolate the contribution of each component of the proposal.

- *MCP, hierarchical* and *SCAD, hierarchical*: the proposed estimator (2.5).
- *Lasso, hierarchical*: the same model and hierarchy with the convex lasso penalty; this is the Cox analogue of the decompositions of Cheng et al. (2015) and Gross and Tibshirani (2016) and isolates the effect of the penalty.
- *MCP, stratified common effects*: (2.5) with all deviations fixed at zero; it isolates the effect of modelling heterogeneity.
- *MCP, SCAD, lasso and elastic net (mixing 0.5) with naive pooling*: a single unstratified Cox model for the stacked data, fitted with *ncvreg* (Breheny and Huang, 2011) or *glmnet* (Friedman et al., 2010); comparison with the stratified common-effect fit isolates the effect of stratification.

All methods receive the same within-study standardised covariates and the same five study- and event-stratified folds. The hierarchical and stratified methods are tuned as in Section 3.2; the pooled methods use the cross-validated partial-likelihood deviance of their packages over their default one-dimensional paths. For every method we record both the minimum and the one-standard-error choice.

5.3 Performance measures

Global selection is assessed against $\mathcal{A}_0$ ($s_\alpha = 6$) through the true positive rate $\text{TPR} = \text{TP}/s_\alpha$, the false discovery rate $\text{FDR} = \text{FP}/|\hat{\mathcal{A}}|$ (zero if nothing is selected), the Matthews correlation coefficient $\text{MCC} =$

$(TP \cdot TN - FP \cdot FN) / \{(TP + FP)(TP + FN)(TN + FP)(TN + FN)\}^{1/2}$, which remains informative when $s_\alpha \ll p$ (Matthews, 1975; Chicco and Jurman, 2023), and the proportion of replicates with exact recovery $\hat{\mathcal{A}} = \mathcal{A}_0$. Estimation of the study-specific effects is assessed by $\text{MSE}(\boldsymbol{\theta}) = (Kp)^{-1} \sum_{k,j} (\hat{\theta}_{kj} - \theta_{kj})^2$. Recovery of heterogeneity is assessed by the true positive and false discovery rates of the nonzero deviations, TPR_{dev} and FDR_{dev} , computed over the Kp pairs (k, j) ; these are defined only for the hierarchical methods. For the proposed MCP estimator we also record, for each truly active covariate that is selected, whether the 95% post-selection Wald interval of Section 3.3 covers α_{0j} . Results are Monte Carlo means; error bars in the figures are ± 1.96 Monte Carlo standard errors, which are also given in the repository.

5.4 Results

Table 2 gives the base scenario, Figures 1 to 4 summarise all scenarios, and Table A1 in the appendix lists the main metrics for every scenario and method. Monte Carlo standard errors of the MCC of hierarchical MCP are between 0.008 and 0.032. Five findings stand out.

Stratification matters. The stratified common-effect MCP estimator has a higher MCC than MCP with naive pooling in all 13 scenarios under the one-standard-error rule (for example 0.949 against 0.884 in the base scenario, and 0.882 against 0.766 with strong heterogeneity). Ignoring the study structure of the risk sets therefore costs selection accuracy even though every method receives the same within-study standardised covariates.

Nonconvex penalties are necessary for sparse selection. Whatever the model structure, the lasso and the elastic net select far too many covariates: under the minimum rule their false discovery rate is 0.64–0.88 in every scenario, and the hierarchical lasso behaves like the pooled lasso. Under the same rule the MCP estimators have false discovery rates of 0.06–0.44. SCAD is consistently less sparse than MCP, in agreement with its wider transition region.

Global selection: the hierarchy helps under the one-standard-error rule. With the one-standard-error rule, hierarchical MCP has the highest MCC in 8 of the 13 scenarios (one of them a tie): the base scenario (0.972; exact recovery in 76% of replicates), no heterogeneity, strong heterogeneity (0.933 against 0.882 for the common-effect model), strong correlation, weak signals, ten studies, unequal study sizes and $p = 500$ (0.902 against 0.879). It is within 0.006 of the best in four further scenarios (independent covariates, heavy censoring, where hierarchical SCAD is best, $p = 200$, and small studies), and it is clearly outperformed only in the dense-heterogeneity scenario (0.893 against 0.912), where the sharing rule is violated by construction. With the minimum rule the ordering reverses: the common-effect model has the higher MCC in 11 of 13 scenarios, by 0.003–0.05 (0.11 in the dense-heterogeneity scenario), because the extra flexibility of the deviations lets cross-validation choose a smaller λ_α and admit a few more false global effects. This is the basis for the recommendation in Section 3.2: use the one-standard-error rule for selection.

Study-specific effects: the hierarchy helps substantially. The hierarchical estimators have the lowest $\text{MSE}(\boldsymbol{\theta})$ in every scenario with heterogeneity except small studies with $p > n_k$ (Figure 3). Under the minimum rule, the error of the common-effect MCP estimator is 1.2 to 3.2 times that of hierarchical MCP in these scenarios and 7.7 times with strong heterogeneity. When there is no heterogeneity the two are equal (ratio 0.97), and hierarchical MCP selects essentially no spurious deviations (0.04 per replicate under the minimum rule and none under the one-standard-error rule), so modelling heterogeneity that is absent costs nothing. The same holds for small studies with $p > n_k$ (ratio 0.98), where the data carry too little study-level information to estimate deviations.

Which deviations are found. Deviation detection is conservative (Figure 4). Under the one-standard-error rule hierarchical MCP finds 8% to 39% of the true deviations with a false discovery rate of at most 0.20; under the minimum rule it finds 19% to 52% with a false discovery rate of 0.09 to 0.47. The deviations

missed are the small ones: with $\sigma_\varepsilon = 0.4$, half of the true deviations are smaller than 0.27 in absolute value, which cannot be distinguished from zero with 200 patients per study; with $\sigma_\varepsilon = 0.8$ the detection rate rises to 0.39 (one-standard-error rule) and 0.52 (minimum rule). A selected deviation is thus usually real, but the absence of a selected deviation is weak evidence of homogeneity.

Coverage of the Wald intervals. Table 3 shows that the post-selection intervals attain nominal coverage when the model is correctly selected: 0.94 in the scenario without heterogeneity (minimum rule). With heterogeneity, coverage drops to 0.69–0.82 and is lowest for the strong effects, which are precisely covariates 1–4, the ones carrying deviations. The cause is the conservative deviation detection just described: when a true deviation is not selected, the refit estimates α_j as an average over sharing and deviating studies, which is biased for the shared effect. Coverage is also poor when selection itself is unreliable (strong correlation, weak signals, small studies). These intervals should therefore be reported together with the bootstrap selection frequencies, as we do in Section 6.

Table 2: Base scenario ($K = 5$, $n_k = 200$, $p = 100$), Monte Carlo means over $R = 25$ replicates. TPR, FDR, MCC and Exact refer to the global active set; $|\hat{\mathcal{A}}|$ is the number of selected covariates; TPR_{dev} and FDR_{dev} refer to the nonzero deviations (hierarchical methods only); Time is the cross-validation time on one core. Bold: best value within rule.

Method	TPR	FDR	MCC	Exact	$ \hat{\mathcal{A}} $	$\text{MSE}(\theta) \times 100$	TPR_{dev}	FDR_{dev}	Time (s)
<i>One-standard-error rule</i>									
MCP, hierarchical (proposed)	1.000	0.049	0.972	0.76	6.44	0.136	0.170	0.070	4.49
SCAD, hierarchical (proposed)	0.980	0.131	0.910	0.44	7.24	0.154	0.285	0.146	3.42
Lasso, hierarchical	1.000	0.539	0.635	0.04	15.08	0.460	0.405	0.360	2.09
MCP, stratified common effects	0.940	0.027	0.949	0.68	5.84	0.320	–	–	0.57
MCP, naive pooling	0.800	0.000	0.884	0.36	4.80	0.598	–	–	0.79
SCAD, naive pooling	0.860	0.044	0.896	0.32	5.44	0.644	–	–	0.57
Lasso, naive pooling	0.980	0.421	0.724	0.04	11.08	0.859	–	–	0.59
Elastic net, naive pooling	0.993	0.563	0.621	0.00	14.92	0.945	–	–	0.64
<i>Minimum-deviance rule</i>									
MCP, hierarchical (proposed)	1.000	0.137	0.920	0.44	7.24	0.094	0.315	0.181	4.49
SCAD, hierarchical (proposed)	1.000	0.325	0.799	0.04	9.76	0.091	0.395	0.334	3.42
Lasso, hierarchical	1.000	0.780	0.403	0.00	29.80	0.250	0.565	0.737	2.09
MCP, stratified common effects	0.987	0.082	0.946	0.56	6.56	0.268	–	–	0.57
MCP, naive pooling	0.987	0.189	0.883	0.28	7.60	0.299	–	–	0.79
SCAD, naive pooling	0.993	0.387	0.756	0.04	10.44	0.299	–	–	0.57
Lasso, naive pooling	1.000	0.749	0.446	0.00	25.04	0.558	–	–	0.59
Elastic net, naive pooling	1.000	0.812	0.365	0.00	33.28	0.613	–	–	0.64

Table 3: Hierarchical MCP: proportion of truly active covariates selected (TPR) and empirical coverage of nominal 95% post-selection Wald intervals for α_{0j} among selected active covariates, overall and for $|\alpha_{0j}| \geq 0.5$.

Scenario	One-SE rule			Minimum rule		
	TPR	Cov. (all)	Cov. ($ \alpha_j \geq 0.5$)	TPR	Cov. (all)	Cov. ($ \alpha_j \geq 0.5$)
Base	1.000	0.813	0.730	1.000	0.820	0.760
No heterogeneity	0.973	0.884	0.900	1.000	0.940	0.940
Strong heterogeneity	0.980	0.694	0.667	1.000	0.800	0.730
Heavy censoring	0.793	0.689	0.621	0.947	0.782	0.717
Weak signals	0.600	0.644	0.560	0.740	0.739	0.720
$p = 500$	0.893	0.687	0.646	0.987	0.750	0.687
Small studies, $p > n_k$	0.693	0.596	0.562	0.793	0.605	0.582

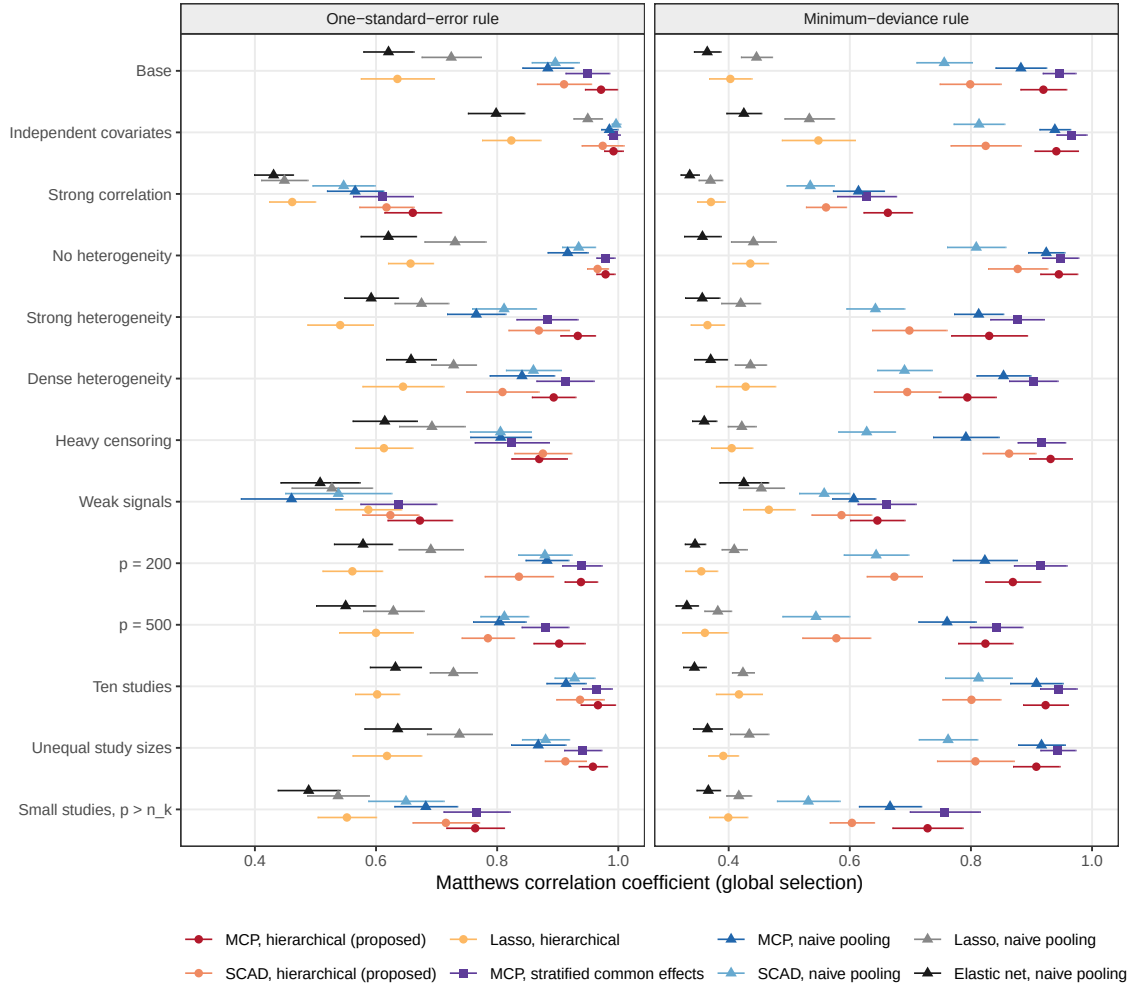


Figure 1: Matthews correlation coefficient of global selection in all scenarios, under the one-standard-error rule (left) and the minimum rule (right). Bars: ± 1.96 Monte Carlo standard errors.

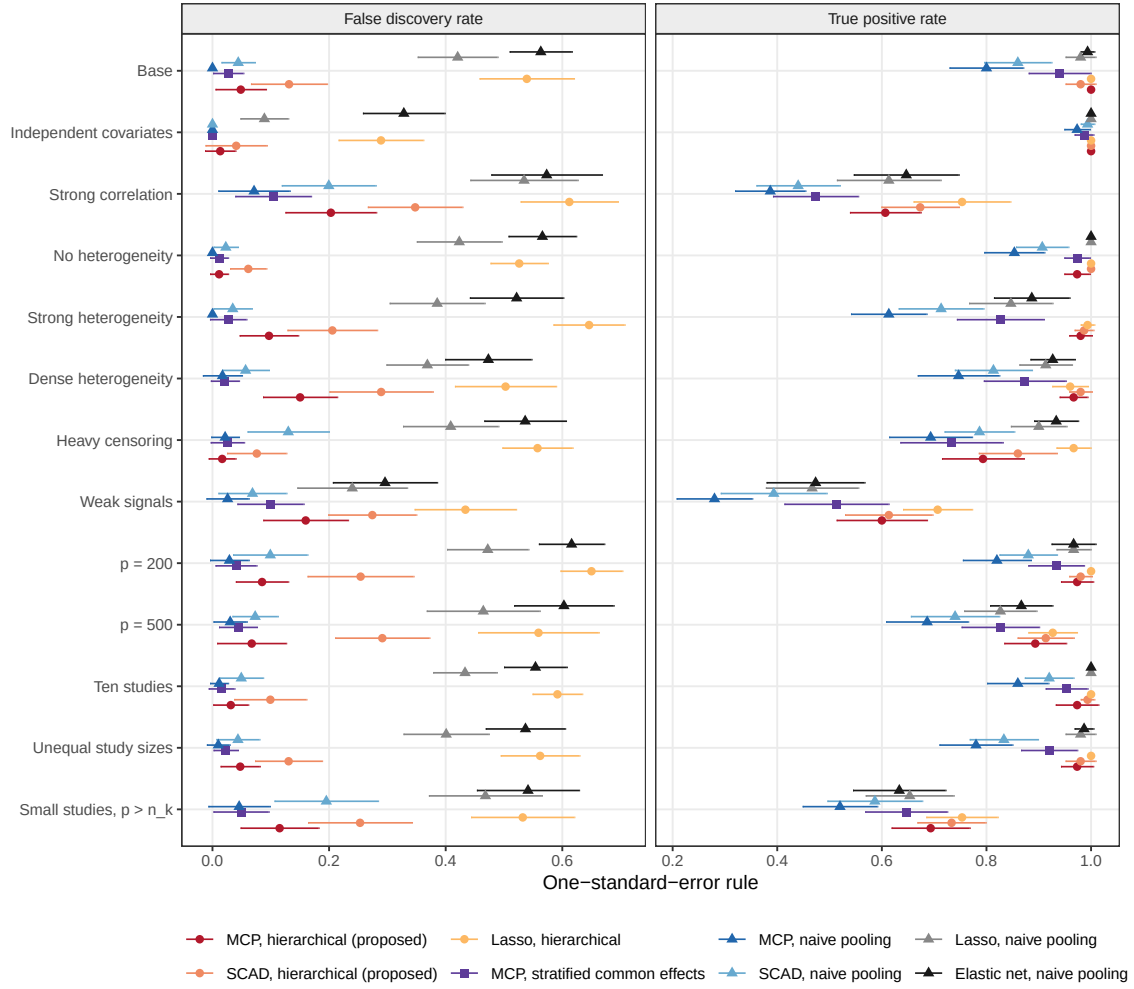


Figure 2: False discovery rate and true positive rate of global selection (one-standard-error rule).

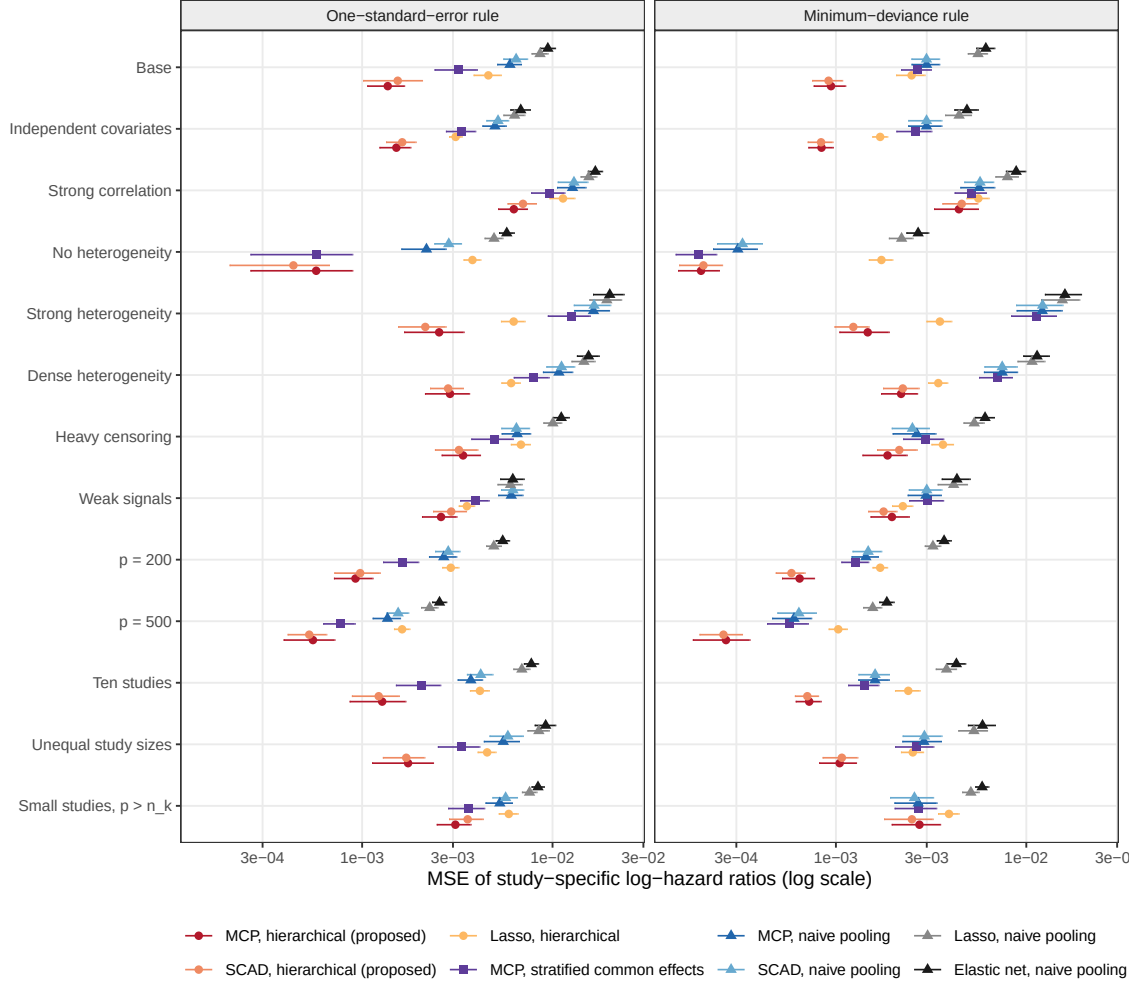


Figure 3: Mean squared error of the study-specific log-hazard ratios $\hat{\theta}_{kj}$ (log scale).

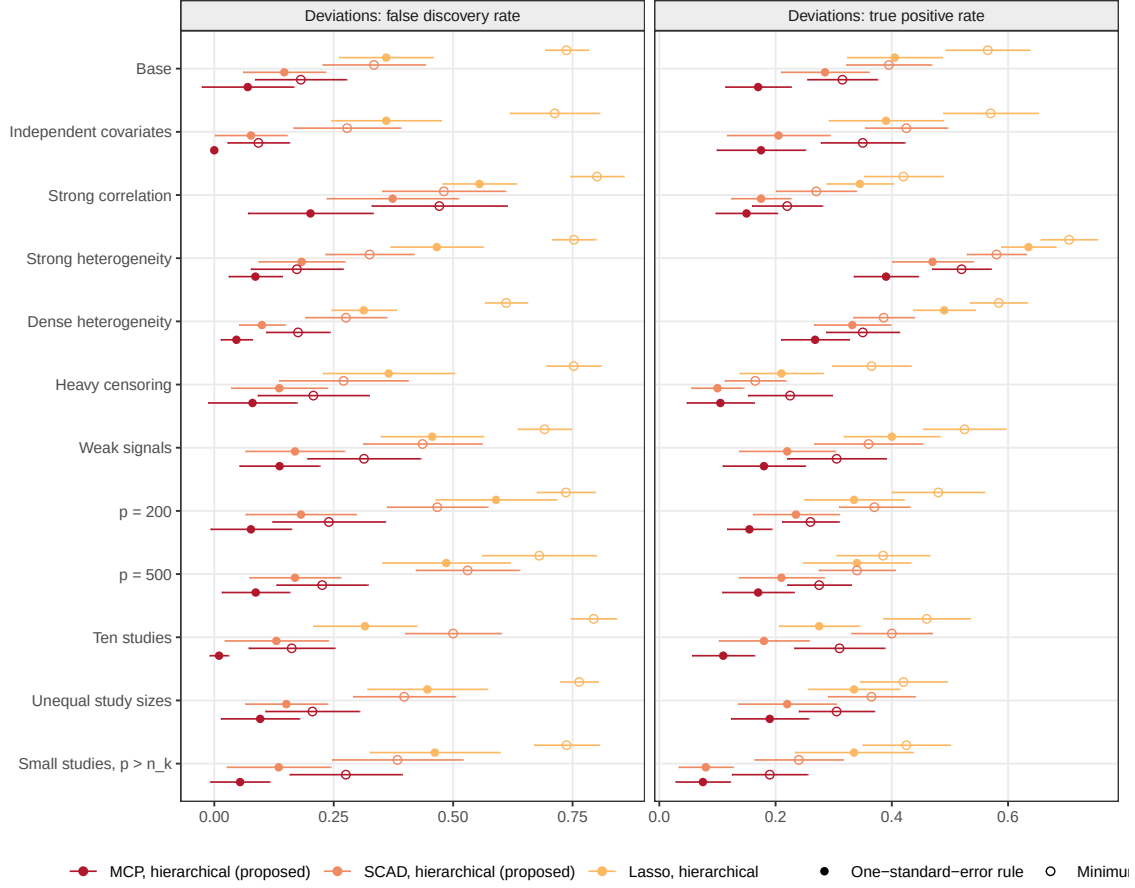


Figure 4: Recovery of the nonzero deviations by the three hierarchical estimators: true positive rate and false discovery rate over the Kp pairs (k, j) . Filled: one-standard-error rule; open: minimum rule. The scenario without heterogeneity is omitted because it has no true deviations.

6 Application: six ovarian cancer cohorts

6.1 Data and pre-processing

We analysed overall survival in six cohorts of the Bioconductor package `curatedOvarianData` (Ganzfried et al., 2013), which provides harmonised gene-expression profiles and clinical annotation for published ovarian cancer studies and has been used for cross-study validation of prognostic signatures (Waldron et al., 2014; Bernau et al., 2014). We used the six cohorts analysed in the first version of this paper: all participants with a recorded vital status and positive follow-up were retained, giving $N = 1,446$ patients and 735 deaths (Table 4). The cohorts differ markedly in size (58 to 557 patients), censoring (30% to 59%) and follow-up, and their Kaplan–Meier curves cross (Figure 5), so a common baseline hazard is not tenable and stratification by study is required.

Of the 12,249 genes measured in all six cohorts, we kept $p = 500$ with an outcome-blind filter: genes were ranked by their interquartile range within each cohort, and the 500 with the best average rank were retained. The filter does not use survival, so it does not bias the subsequent selection. With $p = 500$, the number of candidate genes exceeds the sample size of four of the six cohorts and is 68% of the total number of deaths. Each gene was standardised within cohort, so hazard ratios are per within-cohort standard deviation. The first version of this paper intersected the 2,000 most variable genes of each cohort and obtained only $p = 66$; the present filter gives a genuinely high-dimensional problem and a gene set that does not depend on the platform-specific variability ranking of any single cohort.

Table 4: Cohorts from `curatedOvarianData`. Median overall survival and median follow-up (reverse Kaplan–Meier) are in years.

Study	GEO/TCGA series	n_k	Deaths	Censored (%)	Median OS (yr)	Median follow-up (yr)
A	TCGA	557	290	48	3.70	4.30
B	GSE32062.GPL6480	260	121	53	4.90	4.70
C	GSE9891	276	113	59	3.90	3.00
D	GSE26712	185	129	30	3.80	7.50
E	GSE17260	110	46	58	4.40	3.90
F	GSE30161	58	36	38	4.20	6.90
Total		1,446	735	49		

6.2 Analysis

All eight methods of Section 5.2 were applied with the same five study- and event-stratified folds (seed 2026) and the settings of Section 3.2. For the proposed MCP estimator we report the model chosen by the minimum rule, whose purpose is to estimate study-specific effects, and we mark which of its genes are also kept by the one-standard-error rule. For this model we computed post-selection Wald intervals and p -values (Section 3.3), Holm-adjusted p -values across the selected genes, and selection frequencies in 100 within-study bootstrap resamples refitted at the selected $(\lambda_\alpha, \lambda_\epsilon)$. Three sensitivity analyses were carried out: (i) the number of folds, $V = 10$ instead of 5; (ii) refitting the selected genes in a Cox model with a shared gamma frailty for study instead of stratification, and in a naively pooled Cox model; and (iii) cross-study validation, in which each cohort in turn is held out, every method is tuned and fitted on the other five, and Harrell’s concordance index (Harrell et al., 1996) of the global risk score $\mathbf{x}^\top \hat{\boldsymbol{\alpha}}$ is computed in the held-out cohort (Bernau et al., 2014).

6.3 Results

Tuning and model size. Figure 6 shows the cross-validated deviance of the hierarchical MCP model over the two-dimensional grid. The minimum is attained at $r = 1$ and $\lambda_\alpha = 0.056$. The curve is flat near its minimum: the one standard error of the deviance (16.2) spans all ratios $r \geq 1$ over a range of λ_α , so the one-standard-error rule moves to the larger $\lambda_\alpha = 0.089$ with $r = 2$, a two-gene model. The path is truncated

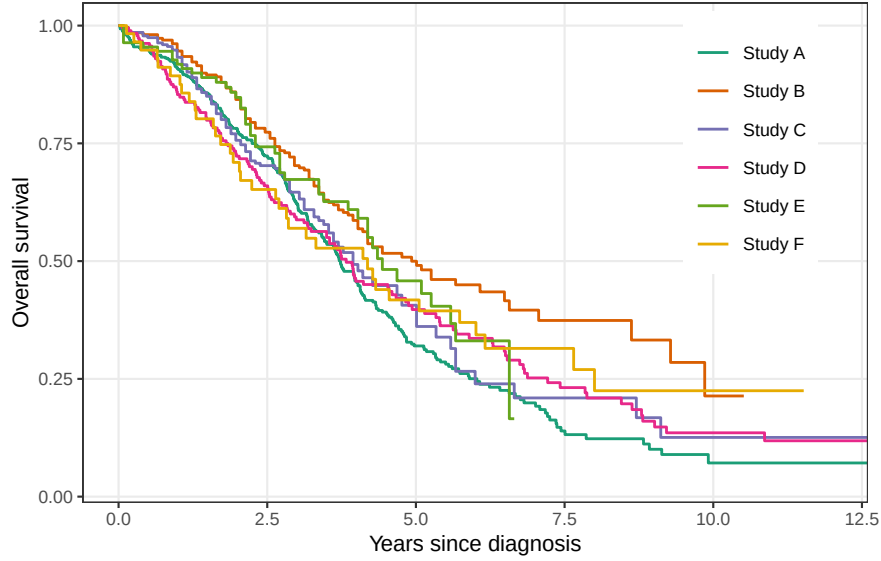


Figure 5: Kaplan–Meier estimates of overall survival by cohort. The curves cross, which argues against a common or proportional baseline hazard.

($d_{\max} = 100$) well to the right of both choices, so the grid does not constrain the selection. The number of genes selected by each method is in Table 5. The nonconvex methods select small models (hierarchical MCP 6 genes, stratified common-effect MCP 4, MCP with naive pooling 8 under the minimum rule), whereas the lasso and the elastic net with naive pooling select dozens of genes, in line with the false discovery rates observed in Section 5.

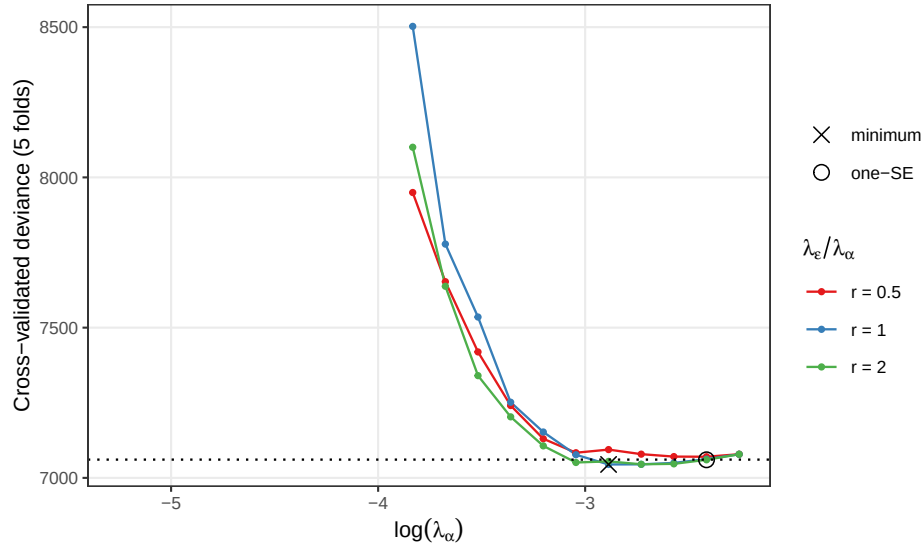


Figure 6: Five-fold cross-validated deviance of the hierarchical MCP model as a function of λ_α for the three ratios $r = \lambda_\varepsilon/\lambda_\alpha$. Cross: minimum rule; circle: one-standard-error rule; dotted line: minimum plus one standard error.

Selected genes and their effects. Table 6 and Figure 7 report the six genes selected by the hierarchical MCP model. Two genes are strongly supported: *GMPT* (HR 0.82, 95% CI 0.76–0.89, Holm-adjusted $p = 3.3 \times 10^{-6}$) and *CXCL9* (HR 0.84, 0.78–0.90, $p_{\text{Holm}} = 1.5 \times 10^{-5}$) are associated with longer survival, and

Table 5: Number of genes (out of $p = 500$) and of nonzero deviations selected by each method under the minimum (min) and one-standard-error (1-SE) rules, and the time for five-fold cross-validation.

Method	Genes selected		Deviations		CV time (s)
	min	1-SE	min	1-SE	
MCP, hierarchical (proposed)	6	2	1	0	18.5
SCAD, hierarchical (proposed)	8	7	0	0	10.4
Lasso, hierarchical	42	23	1	0	4.6
MCP, stratified common effects	4	2	–	–	4.4
MCP, naive pooling	8	4	–	–	12.7
SCAD, naive pooling	35	4	–	–	6.3
Lasso, naive pooling	56	11	–	–	23.8
Elastic net, naive pooling	59	13	–	–	24.8

both are selected by all eight methods. Higher expression of *CRYAB* (HR 1.16, 1.08–1.25) and *OSR2* (HR 1.12, 1.04–1.22) is associated with shorter survival, and both remain significant after Holm adjustment; *GAS1* (HR 1.12, 1.03–1.22, $p_{\text{Holm}} = 0.017$) is weaker. *CXCL9* is a chemokine associated with tumour-infiltrating T cells, and its favourable direction is consistent with the known prognostic value of immune infiltration in high-grade serous ovarian cancer; we do not attempt further biological interpretation.

The sixth gene, *IGF2*, illustrates what the hierarchical model adds. Its global effect is close to null (penalised HR 1.01; refitted HR 1.08, 95% CI 0.99–1.17, $p = 0.085$), but the model selects a deviation in Study D (GSE26712), where the hazard ratio is 1.33 (1.12–1.58). The association of *IGF2* with survival is thus concentrated in one cohort. A pooled or common-effect analysis can only report the averaged hazard ratio (1.12–1.13 in Table 7), which is significant but misleading about how general the effect is; the first version of this paper reported *IGF2* as the strongest global signal for exactly this reason. Note that the penalised $\exp(\hat{\alpha}_j)$ and the refitted HR differ most for the weakest genes, where the MCP penalty is still in its shrinking region.

Table 6: Genes selected by hierarchical MCP (minimum rule). $\exp(\hat{\alpha}_j)$: penalised global hazard ratio per within-cohort SD. HR, 95% CI and p : Wald inference from the unpenalised stratified refit on the selected support (Section 3.3); p_{Holm} : Holm-adjusted across the six genes. Dev.: number of cohorts with a nonzero deviation. Boot. freq.: selection frequency in 100 within-study bootstrap resamples. Methods: number of the eight methods selecting the gene. *Also selected under the one-standard-error rule.

Gene	$\exp(\hat{\alpha}_j)$	HR	95% CI	p	p_{Holm}	Dev.	Boot. freq.	Methods
<i>GMPT*</i>	0.80	0.82	(0.76, 0.89)	5.5e-07	3.3e-06	0	0.49	8/8
<i>CXCL9</i>	0.84	0.84	(0.78, 0.90)	2.9e-06	1.5e-05	0	0.65	8/8
<i>CRYAB</i>	1.17	1.16	(1.08, 1.25)	0.0001	0.0004	0	0.26	8/8
<i>OSR2</i>	1.10	1.12	(1.04, 1.22)	0.0032	0.0095	0	0.31	8/8
<i>GAS1</i>	1.02	1.12	(1.03, 1.22)	0.0086	0.0172	0	0.19	7/8
<i>IGF2</i>	1.01	1.08	(0.99, 1.17)	0.0853	0.0853	1	0.32	7/8

Stability and sensitivity. Selection is only moderately stable (Figure 8). *CXCL9* is selected in 65% of bootstrap resamples and *GMPT* in 49%; the other selected genes in 19%–32%. With $V = 10$ folds, the minimum rule selects *GMPT* and *MFAP4*, and the one-standard-error rule selects no gene; with $V = 5$ the one-standard-error rule also selects *GMPT* and *MFAP4*. *GMPT* is therefore the one gene selected under every tuning choice that selects any gene, whereas the membership of the other genes depends on the tuning; the evidence for a prognostic signal beyond *GMPT* and *CXCL9* is modest. The hazard ratios of the selected genes are practically identical when study is handled by stratification, by a shared gamma frailty, or ignored (Table 7); the estimated frailty variance is 0.018, so baseline differences between cohorts are small on the scale

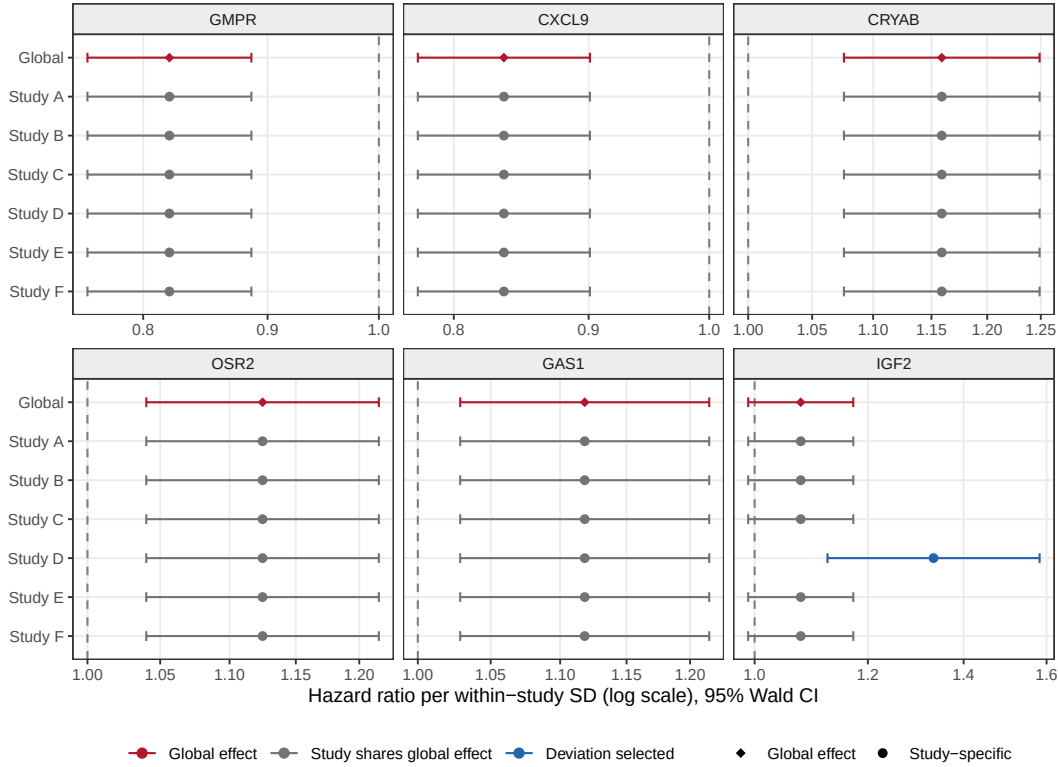


Figure 7: Global (red diamond) and study-specific (circles) hazard ratios with 95% Wald intervals for the six genes selected by hierarchical MCP. Blue: cohort with a selected deviation; grey: cohort sharing the global effect, whose interval is therefore that of the global effect.

of these gene effects and do not confound them. Stratification remains preferable because it does not require proportional baselines, which Figure 5 contradicts.

Table 7: Hazard ratio (95% CI) per within-cohort SD of the selected genes when refitted jointly under three ways of handling the cohorts.

Gene	Stratified Cox	Shared gamma frailty	Naive pooling
<i>GMPR</i>	0.82 (0.76, 0.89)	0.83 (0.76, 0.89)	0.83 (0.77, 0.90)
<i>CXCL9</i>	0.84 (0.78, 0.90)	0.83 (0.77, 0.89)	0.83 (0.77, 0.89)
<i>CRYAB</i>	1.16 (1.08, 1.25)	1.16 (1.08, 1.25)	1.16 (1.08, 1.25)
<i>OSR2</i>	1.13 (1.05, 1.22)	1.13 (1.05, 1.23)	1.14 (1.05, 1.23)
<i>GAS1</i>	1.11 (1.02, 1.21)	1.11 (1.02, 1.20)	1.11 (1.02, 1.20)
<i>IGF2</i>	1.12 (1.04, 1.21)	1.13 (1.04, 1.22)	1.13 (1.05, 1.22)

Cross-study validation. Table 8 reports the concordance of the global risk score in each held-out cohort. All values are modest (0.51–0.60), in line with earlier cross-study evaluations of prognostic signatures in these data (Waldron et al., 2014; Bernau et al., 2014). The dense models predict best: the elastic net and the lasso with naive pooling reach a mean C of 0.60 with about 40 genes, against 0.56 for hierarchical MCP with about five genes (minimum rule). Under the one-standard-error rule the sparse models often select no gene at all in the training cohorts, and their C is close to 0.5. Hierarchical and common-effect MCP give identical held-out predictions here, because the only deviation selected does not alter the global effects used for prediction. The sparse models therefore give up some predictive accuracy for a short, interpretable list of

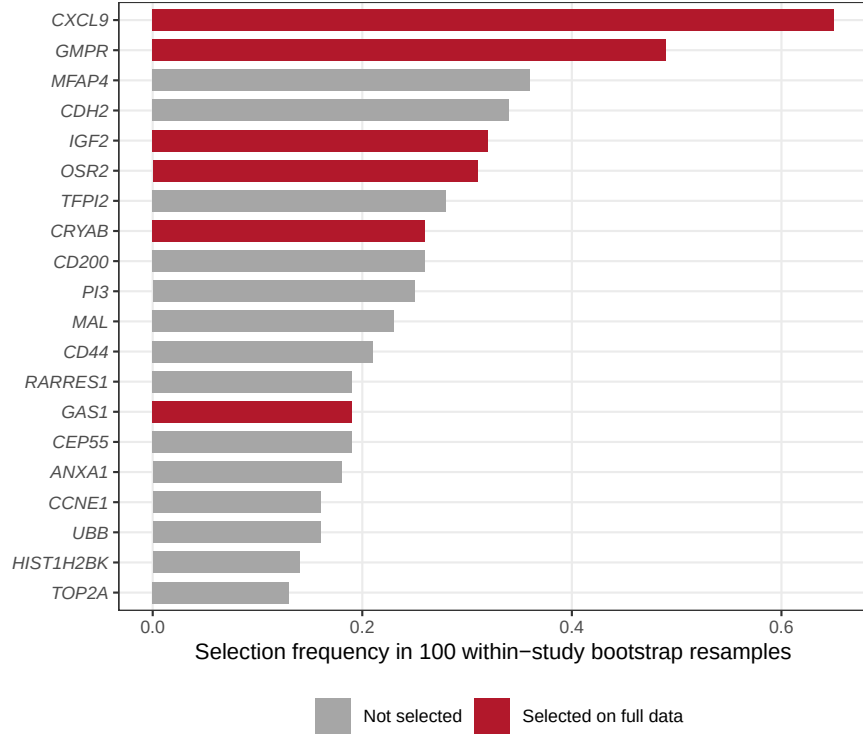


Figure 8: The 20 genes most often selected by hierarchical MCP in 100 within-study bootstrap resamples at the tuning parameters chosen on the full data. Red: genes selected on the full data.

genes; when prediction in new cohorts is the goal, a dense model is preferable.

Table 8: Leave-one-study-out validation: Harrell’s C of the global risk score $\mathbf{x}^\top \hat{\boldsymbol{\alpha}}$ in the held-out cohort, for each method tuned and fitted on the other five cohorts, and the mean number of genes selected.

	MCP-H	SCAD-H	MCP-S	MCP-P	LASSO-P	ENet-P
<i>Minimum rule</i>						
Study A held out	0.549	0.565	0.549	0.595	0.598	0.597
Study B held out	0.542	0.549	0.542	0.562	0.583	0.581
Study C held out	0.605	0.610	0.605	0.610	0.656	0.656
Study D held out	0.587	0.637	0.587	0.611	0.607	0.611
Study E held out	0.579	0.590	0.579	0.592	0.600	0.598
Study F held out	0.505	0.511	0.505	0.562	0.572	0.580
Mean C	0.561	0.577	0.561	0.589	0.603	0.604
Mean no. of genes	4.7	9.2	4.7	9.8	38.3	42.7
<i>One-standard-error rule</i>						
Study A held out	0.500	0.500	0.500	0.551	0.575	0.577
Study B held out	0.500	0.565	0.500	0.558	0.564	0.564
Study C held out	0.603	0.612	0.603	0.615	0.606	0.607
Study D held out	0.500	0.500	0.500	0.500	0.638	0.640
Study E held out	0.500	0.567	0.500	0.547	0.546	0.549
Study F held out	0.465	0.528	0.465	0.465	0.555	0.561
Mean C	0.511	0.545	0.511	0.539	0.581	0.583
Mean no. of genes	0.5	2.3	0.5	1.8	9.5	10.0

6.4 Conclusions from the application

The analysis supports four conclusions. First, among 500 candidate genes, *GMPT* and *CXCL9* are associated with longer overall survival (hazard ratios 0.82 and 0.84 per SD, Holm-adjusted $p < 10^{-4}$), selected by every method and the most stable under resampling; *CRYAB* and *OSR2* are associated with shorter survival but are less stable. Second, these effects are common to all six cohorts: no deviation was selected for them, and the frailty and stratified analyses agree. Third, the association of *IGF2* with survival is confined to one cohort (Study D); this is the kind of finding that pooled or common-effect analyses cannot make and that the hierarchical decomposition was designed to reveal. Fourth, the prognostic signal in these expression data is weak and diffuse: selection beyond the two leading genes depends on the tuning rule and the number of folds, and dense models predict new cohorts somewhat better than any sparse model. These conclusions are about genes; a clinical model would add stage, grade and residual disease.

7 Discussion

We proposed a one-stage stratified Cox model for IPD meta-analysis in which each study-specific log-hazard ratio is the sum of a global effect and a sparse, fixed study-specific deviation, estimated by nonconvex penalised partial likelihood under a hierarchy constraint and a sharing rule that identifies the decomposition. Compared with the first version of this work, the objective function is now stated explicitly, the zero-sum constraint has been replaced by an identification rule compatible with sparse deviations, the two tuning parameters are chosen by a fully specified two-dimensional cross-validation, and the method is available as an R package with compiled code, tests, and scripts that regenerate every result.

What the hierarchy adds. The simulation separates three ingredients. Stratification of the partial likelihood improves global selection in every scenario; nonconvex penalties are necessary to control false discoveries, whether or not the hierarchy is used; and the hierarchical deviations mainly improve the estimation of study-specific effects, by a factor of 1.2 to 7.7 in squared error when heterogeneity is present and at no cost when it is absent. With the one-standard-error rule the hierarchical MCP estimator also gives the most accurate global selection in most scenarios. The first version of this paper attributed the lower false discovery rate to the decomposition; that comparison did not include a stratified common-effect competitor and was based on an unstratified implementation, and we no longer make that claim. In the application the hierarchy made one substantive difference: it showed that the *IGF2* association is confined to one cohort.

Relation to frailty models. Stratification and fixed deviations are an alternative, not a replacement, to frailty and random-slope models (Vaida and Xu, 2000; Ripatti and Palmgren, 2000; Groll et al., 2017; Utazirubanda et al., 2021). When the number of studies is large and the question is the size of the between-study variance, a random-effects formulation is natural. With the handful of studies typical of IPD meta-analyses, many candidate covariates, and the question of *which* studies depart from a common effect, the fixed-deviation formulation avoids a distributional assumption that cannot be checked and remains a penalised partial likelihood. In our application the two formulations gave the same hazard ratios for the selected genes.

Limitations. First, the Wald intervals are conditional on the selected model and under-cover in finite samples (Table 3); they should be read with the bootstrap selection frequencies, and selective-inference methods (Berk et al., 2013) for the stratified Cox model with a hierarchy constraint remain to be developed. Second, the hierarchy constraint cannot represent a covariate that acts only in a minority of studies other than through a small global effect with large deviations; this is transparent in the output but may be undesirable when such covariates are expected. Third, our simulation uses $R = 25$ replicates per scenario because of computational limits; Monte Carlo standard errors are reported, and the differences we emphasise exceed them. Fourth, the theory covers $p = O(N^c)$ with $c < 1$; for $p \gg N$ a screening step (Fan and Lv, 2008; Fan et al., 2010) would be needed before penalisation. Fifth, the application uses gene expression only; a clinical

analysis would adjust for stage, grade and residual disease, which are incompletely recorded across these cohorts.

Extensions. Natural extensions are an information-criterion alternative to cross-validation such as the extended BIC, group penalties for sets of covariates such as pathways, time-varying deviations, and studies that measure different subsets of covariates.

Code and data availability

All code is openly available at <https://github.com/EmmanuelMasavoDjegou/Hierarchical-Variable-Selection-in-Penalized-Cox-Models-for-IPD-Meta-Analysis> (release v1.0.0). The repository contains the R package `hiermetacox`; the scripts `simulation/01_run_simulation.R` and `simulation/02_make_tables_figures.R`, which generate every simulation table and figure in this paper; the scripts `application/01_prepare_data.R-03_make_tables_figures.R`, which generate every table and figure of Section 6; all per-replicate simulation results; and a consistency check between code and manuscript. Tables in this paper are \input directly from files written by these scripts. The ovarian cancer data are publicly available in the Bioconductor package `curatedOvarianData` (Ganzfried et al., 2013). All computations used R 4.3.3 (R Core Team, 2024).

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A Proofs

A.1 Set-up

We work with the free parameter $\boldsymbol{\vartheta} \in \mathbb{R}^{(K+1)p_N}$ of Section 4. Write $\ell_N(\boldsymbol{\vartheta}) = N^{-1} \sum_k \ell_k(\boldsymbol{\alpha} + \boldsymbol{\varepsilon}_k)$, so that $\mathcal{Q}_\lambda(\boldsymbol{\vartheta}) = -\ell_N(\boldsymbol{\vartheta}) + \mathcal{P}(\boldsymbol{\vartheta})$ with $\mathcal{P}$ the penalty in (2.6), and denote by $\dot{\ell}_N$ and $\ddot{\ell}_N$ its gradient and Hessian. The coordinate of $\boldsymbol{\vartheta}$ corresponding to α_j has design column $\mathbf{x}_{\cdot j}$; the coordinate corresponding to ε_{kj} has column $\mathbf{x}_{\cdot j} \mathbf{1}(\text{study} = k)$. The weights ω_k converge to positive constants, so the deviation penalties $P(\omega_k |\cdot|; \lambda_\varepsilon, \gamma)$ satisfy Assumption 6 with λ_ε replaced by $\lambda_\varepsilon/\omega_k$, which is of the same order; we suppress ω_k below. For a vector $\mathbf{v}$ we write $\mathbf{v}_S$ for its subvector on S .

A.2 Auxiliary lemmas

Lemma A.1 (Score). *Under Assumptions 2–4 and 8, $\|\dot{\ell}_N(\boldsymbol{\vartheta}_0)\|_\infty = O_P(\sqrt{\log p_N/N})$ and $\|\dot{\ell}_{N,S_0}(\boldsymbol{\vartheta}_0)\|_2 = O_P(\sqrt{s_0/N})$.*

Proof. In counting-process notation each coordinate of the score at the truth is

$$\dot{\ell}_{N,j}(\boldsymbol{\vartheta}_0) = N^{-1} \sum_k \int_0^{\tau_k} \phi_{kj}(t) dM_k(t),$$

with M_k the mean-zero martingale of study k and ϕ_{kj} the predictable covariate process of column j centred at its risk-set average *within study* k ; for a deviation coordinate of study k' , $\phi_{kj} \equiv 0$ for $k \neq k'$. By Assumption 4, $|\phi_{kj}| \leq 2M$, and by Assumption 3 the compensator of M_k is $O(n_k)$, so the predictable variation of $\dot{\ell}_{N,j}(\boldsymbol{\vartheta}_0)$ is $O(N^{-1})$ uniformly in j . Freedman's inequality for martingales with bounded jumps gives $\Pr(|\dot{\ell}_{N,j}(\boldsymbol{\vartheta}_0)| > a) \leq 2 \exp(-cNa^2)$ for small a and a constant c not depending on j . A union bound over the $(K+1)p_N$ coordinates with $a = M_0 \sqrt{\log p_N/N}$ and $cM_0^2 > 1$ yields the first claim, because K is fixed. For the second, the covariance of $\dot{\ell}_{N,S_0}(\boldsymbol{\vartheta}_0)$ has trace $O(s_0/N)$ by the same bounds, and Markov's inequality applied to its squared norm gives the rate. $\square$

Lemma A.2 (Local expansion). *Under Assumptions 2–5, uniformly over $\mathcal{N}_N(C) = \{\boldsymbol{\vartheta} : \|\boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0\| \leq Cs_0^{1/2}N^{-1/2}, \boldsymbol{\vartheta}_{S_0^c} = \mathbf{0}\}$, $\ell_N(\boldsymbol{\vartheta}) = \ell_N(\boldsymbol{\vartheta}_0) + (\boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0)^\top \dot{\ell}_N(\boldsymbol{\vartheta}_0) + \frac{1}{2}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0)^\top \ddot{\ell}_N(\boldsymbol{\vartheta}_0)(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0) + o_P(s_0/N)$, and with probability tending to one $c_1/2 \leq \lambda_{\min}(-\ddot{\ell}_{N,S_0S_0}(\boldsymbol{\vartheta})) \leq \lambda_{\max}(-\ddot{\ell}_{N,S_0S_0}(\boldsymbol{\vartheta})) \leq 2c_2$ on $\mathcal{N}_N(C)$.*

Proof. Taylor's theorem gives the expansion with remainder $\frac{1}{2}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0)^\top \{\ddot{\ell}_N(\tilde{\boldsymbol{\vartheta}}) - \ddot{\ell}_N(\boldsymbol{\vartheta}_0)\}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0)$. The uniform expansion in Assumption 5 implies $\sup_{\mathcal{N}_N(C)} \|\ddot{\ell}_{N,S_0S_0}(\boldsymbol{\vartheta}) - \ddot{\ell}_{N,S_0S_0}(\boldsymbol{\vartheta}_0)\| = o_P(1)$ and $\|-\ddot{\ell}_{N,S_0S_0}(\boldsymbol{\vartheta}_0) - N^{-1}\mathbf{I}_{N,S_0}(\boldsymbol{\vartheta}_0)\| = o_P(1)$; since $\|\boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0\|^2 = O(s_0/N)$ on $\mathcal{N}_N(C)$ the remainder is $o_P(s_0/N)$, and the eigenvalue bounds follow from Assumption 5 and Weyl's inequality. Assumption 1(ii) is used here: it ensures that the columns indexed by S_0 are not linearly dependent, so that Assumption 5 is not void. $\square$

Lemma A.3 (Penalty). *Under Assumptions 6 and 7, for all large N : $P'(|\alpha_{0j}|; \lambda_\alpha, \gamma) = 0$ for $j \in \mathcal{A}_0$; $P'(|\varepsilon_{0,kj}|; \lambda_\varepsilon, \gamma) = 0$ for $j \in \mathcal{A}_0^{(k)}$; and $P'(0+; \lambda, \gamma) = \lambda$. Moreover $P'(\cdot; \lambda, \gamma)$ vanishes on a neighbourhood of radius $\gg d_N^{1/2}N^{-1/2}$ around each nonzero true coefficient.*

Proof. Immediate from the explicit derivatives of SCAD and MCP, which vanish on $(\gamma\lambda, \infty)$, and from the minimum-signal conditions. $\square$

Lemma A.4 (Oracle minimiser). *Under Assumptions 1–8, with probability tending to one the restriction of $\mathcal{Q}_\lambda$ to $\{\boldsymbol{\vartheta} : \boldsymbol{\vartheta}_{S_0^c} = \mathbf{0}\}$ has a local minimiser $\hat{\boldsymbol{\vartheta}}^\circ$ with $\|\hat{\boldsymbol{\vartheta}}^\circ - \boldsymbol{\vartheta}_0\| = O_P(s_0^{1/2}N^{-1/2})$, and $\hat{\boldsymbol{\vartheta}}^\circ \in \mathcal{H}$.*

Proof. For $\|\mathbf{u}\| = Cs_0^{1/2}N^{-1/2}$ supported on S_0 put $D_N(\mathbf{u}) = \mathcal{Q}_\lambda(\boldsymbol{\vartheta}_0 + \mathbf{u}) - \mathcal{Q}_\lambda(\boldsymbol{\vartheta}_0)$. By Lemma A.2, $D_N(\mathbf{u}) = -\mathbf{u}^\top \dot{\ell}_{N,S_0}(\boldsymbol{\vartheta}_0) - \frac{1}{2}\mathbf{u}^\top \ddot{\ell}_{N,S_0S_0}(\boldsymbol{\vartheta}_0)\mathbf{u} + \Delta_P(\mathbf{u}) + o_P(s_0/N)$. By Cauchy–Schwarz and Lemma A.1, the linear term is $O_P(Cs_0/N)$. The quadratic term is at least $\frac{1}{4}c_1C^2s_0/N$ with probability tending to one. By Lemma A.3 the penalty is locally constant around every nonzero true coefficient, so $\Delta_P(\mathbf{u}) = 0$ for large N .

For C large the quadratic term dominates, so $D_N(\mathbf{u}) > 0$ on the sphere and a local minimiser exists in its interior. Every coordinate in $\mathcal{S}_0$ is bounded away from zero by more than $Cs_0^{1/2}N^{-1/2}$ (Assumption 7), so $\hat{\boldsymbol{\vartheta}}^o$ has exactly the support $\mathcal{S}_0$; by Assumption 1 this support satisfies (2.2) and (2.3), hence $\hat{\boldsymbol{\vartheta}}^o \in \mathcal{H}$. $\square$

A.3 Proof of Theorem 4.1

It suffices to show that $\hat{\boldsymbol{\vartheta}}^o$ of Lemma A.4 is, with probability tending to one, a local minimiser of $\mathcal{Q}_\lambda$ over $\mathcal{H}$. On $\mathcal{S}_0$ it is stationary by construction. For a coordinate $q \notin \mathcal{S}_0$ (an inactive global effect or an inactive deviation), $\hat{\boldsymbol{\vartheta}}^o$ is a local minimiser in direction q if $|\dot{\ell}_{N,q}(\hat{\boldsymbol{\vartheta}}^o)| < P'(0+)$, which equals λ_α or λ_ε . A first-order expansion gives $\dot{\ell}_{N,q}(\hat{\boldsymbol{\vartheta}}^o) = \dot{\ell}_{N,q}(\boldsymbol{\vartheta}_0) + \mathbf{h}_{N,q}^\top(\hat{\boldsymbol{\vartheta}}^o - \boldsymbol{\vartheta}_0) + o_P(N^{-1/2})$ uniformly in q , where $\mathbf{h}_{N,q}$ is the q th row of the Hessian at an intermediate point, bounded in norm by Assumptions 4 and 5. By Lemmas A.1 and A.4, $\max_q |\dot{\ell}_{N,q}(\hat{\boldsymbol{\vartheta}}^o)| = O_P(\sqrt{\log p_N/N} + \sqrt{s_0/N})$, which is $o_P(\min(\lambda_\alpha, \lambda_\varepsilon))$ by Assumptions 7 and 8. Hence all Karush–Kuhn–Tucker inequalities hold strictly with probability tending to one. Because the conditions are strict and the penalty is flat near the nonzero coordinates, small perturbations in any direction that stay in $\mathcal{H}$ do not decrease $\mathcal{Q}_\lambda$, so $\hat{\boldsymbol{\vartheta}}^o$ is a local minimiser over $\mathcal{H}$ and $\hat{\boldsymbol{\vartheta}} = \hat{\boldsymbol{\vartheta}}^o$ has support $\mathcal{S}_0$. This gives both claims. $\square$

A.4 Proof of Theorem 4.2

On the event of Theorem 4.1, $\hat{\boldsymbol{\vartheta}} = \hat{\boldsymbol{\vartheta}}^o$ and, by Lemma A.3, the penalty has zero derivative at $\hat{\boldsymbol{\vartheta}}_{\mathcal{S}_0}$, so $\dot{\ell}_{N,\mathcal{S}_0}(\hat{\boldsymbol{\vartheta}}) = \mathbf{0}$. A Taylor expansion around $\boldsymbol{\vartheta}_0$ gives $\hat{\boldsymbol{\vartheta}}_{\mathcal{S}_0} - \boldsymbol{\vartheta}_{0,\mathcal{S}_0} = -\ddot{\ell}_{N,\mathcal{S}_0\mathcal{S}_0}^{-1}(\boldsymbol{\vartheta}^*)\dot{\ell}_{N,\mathcal{S}_0}(\boldsymbol{\vartheta}_0)$ with $\boldsymbol{\vartheta}^* \in \mathcal{N}_N(C)$, and Lemma A.2 yields $\sqrt{N}(\hat{\boldsymbol{\vartheta}}_{\mathcal{S}_0} - \boldsymbol{\vartheta}_{0,\mathcal{S}_0}) = \mathbf{I}_{\mathcal{S}_0}^{-1}(\boldsymbol{\vartheta}_0)\sqrt{N}\dot{\ell}_{N,\mathcal{S}_0}(\boldsymbol{\vartheta}_0) + o_P(1)$. The active score is a sum over the K independent studies of stochastic integrals of bounded predictable processes against the study martingales, with predictable variation converging to $\mathbf{I}_{\mathcal{S}_0}(\boldsymbol{\vartheta}_0)$; the martingale central limit theorem (Andersen and Gill, 1982) and Slutsky's theorem give the result. The subvectors for $\boldsymbol{\alpha}_{\mathcal{A}_0}$ and for each $\boldsymbol{\varepsilon}_{k,\mathcal{A}_0^{(k)}}$ follow by projection. $\square$

A.5 Proof of Corollary 4.1

By Theorem 4.1 and because K is fixed, $\hat{\mathcal{A}}^{(k)} = \mathcal{A}_0^{(k)} \subseteq \mathcal{A}_0 = \hat{\mathcal{A}}$ for all k simultaneously with probability tending to one, using Assumption 1(i). The second statement is Proposition 4.1(i). $\square$

A.6 Proof of Proposition 4.1

(i) Steps (a)–(d) of the block update in Section 3.1 never produce $\alpha_j = 0$ with a nonzero deviation (step (b)), and step (d) leaves at least one deviation equal to zero for every active covariate; step-halving takes a convex combination of two points of $\mathcal{H}$ and is followed by setting to zero the deviations of any covariate whose global effect has become zero, which does not leave $\mathcal{H}$. (ii) The accept/halve rule accepts a new iterate only if $\mathcal{Q}_\lambda$ has not increased, and otherwise returns the previous iterate. (iii) $\mathcal{Q}_\lambda$ is bounded below because the penalties are nonnegative and the negative log partial likelihood is bounded below by zero, so the nonincreasing sequence converges. If the iterates converge to $\bar{\boldsymbol{\vartheta}}$, the surrogate built at $\bar{\boldsymbol{\vartheta}}$ has the same gradient as $-\ell_N$ at $\bar{\boldsymbol{\vartheta}}$, and $\bar{\boldsymbol{\vartheta}}$ is not moved by an exact coordinate minimisation of the penalised surrogate; for every coordinate whose hierarchy constraint is inactive, exact coordinate optimality of the surrogate at its expansion point is the first-order condition of $\mathcal{Q}_\lambda$ in that coordinate. $\square$

B Additional simulation results

Table A1: Monte Carlo means for all scenarios. $\text{MSE}(\theta)$ is multiplied by 100. Bold: best value within scenario and rule. Monte Carlo standard errors are in `simulation/results/sim_summary.csv`.

Method	One-SE rule				Minimum rule			
	MCC	FDR	Exact	$\text{MSE}(\theta)$	MCC	FDR	Exact	$\text{MSE}(\theta)$
Base ($R = 25$)								
MCP-H	0.972	0.049	0.76	0.136	0.920	0.137	0.44	0.094
SCAD-H	0.910	0.131	0.44	0.154	0.799	0.325	0.04	0.091
LASSO-H	0.635	0.539	0.04	0.460	0.403	0.780	0.00	0.250
MCP-S	0.949	0.027	0.68	0.320	0.946	0.082	0.56	0.268
MCP-P	0.884	0.000	0.36	0.598	0.883	0.189	0.28	0.299
SCAD-P	0.896	0.044	0.32	0.644	0.756	0.387	0.04	0.299
LASSO-P	0.724	0.421	0.04	0.859	0.446	0.749	0.00	0.558
ENet-P	0.621	0.563	0.00	0.945	0.365	0.812	0.00	0.613
Independent covariates ($R = 25$)								
MCP-H	0.992	0.013	0.96	0.151	0.941	0.100	0.60	0.084
SCAD-H	0.974	0.041	0.88	0.162	0.824	0.282	0.24	0.084
LASSO-H	0.823	0.289	0.12	0.310	0.548	0.636	0.04	0.171
MCP-S	0.993	0.000	0.92	0.334	0.966	0.059	0.72	0.263
MCP-P	0.985	0.000	0.84	0.498	0.939	0.108	0.44	0.299
SCAD-P	0.996	0.000	0.96	0.517	0.813	0.308	0.12	0.299
LASSO-P	0.950	0.089	0.52	0.632	0.533	0.663	0.00	0.444
ENet-P	0.798	0.328	0.12	0.681	0.425	0.765	0.00	0.488
Strong correlation ($R = 25$)								
MCP-H	0.661	0.203	0.04	0.626	0.663	0.429	0.00	0.443
SCAD-H	0.617	0.348	0.00	0.699	0.561	0.579	0.00	0.458
LASSO-H	0.462	0.612	0.00	1.134	0.371	0.803	0.00	0.560
MCP-S	0.611	0.104	0.00	0.964	0.628	0.392	0.00	0.517
MCP-P	0.566	0.071	0.00	1.273	0.614	0.357	0.00	0.565
SCAD-P	0.547	0.200	0.00	1.296	0.535	0.535	0.00	0.572
LASSO-P	0.449	0.534	0.00	1.552	0.370	0.802	0.00	0.796
ENet-P	0.431	0.573	0.00	1.678	0.336	0.828	0.00	0.885
No heterogeneity ($R = 25$)								
MCP-H	0.979	0.011	0.76	0.057	0.945	0.095	0.56	0.020
SCAD-H	0.966	0.061	0.60	0.044	0.877	0.203	0.28	0.020
LASSO-H	0.657	0.526	0.00	0.380	0.436	0.756	0.00	0.173
MCP-S	0.979	0.011	0.76	0.057	0.948	0.091	0.56	0.019
MCP-P	0.916	0.000	0.44	0.218	0.924	0.132	0.40	0.031
SCAD-P	0.935	0.023	0.44	0.285	0.809	0.311	0.12	0.032
LASSO-P	0.731	0.423	0.08	0.493	0.441	0.749	0.00	0.221
ENet-P	0.620	0.566	0.00	0.575	0.357	0.815	0.00	0.270
Strong heterogeneity ($R = 25$)								
MCP-H	0.933	0.097	0.44	0.254	0.830	0.269	0.32	0.147
SCAD-H	0.869	0.206	0.28	0.215	0.698	0.459	0.00	0.123
LASSO-H	0.541	0.646	0.00	0.625	0.365	0.810	0.00	0.352
MCP-S	0.882	0.027	0.48	1.257	0.876	0.147	0.36	1.132
MCP-P	0.766	0.000	0.04	1.640	0.813	0.259	0.08	1.213
SCAD-P	0.811	0.035	0.16	1.652	0.642	0.502	0.00	1.217
LASSO-P	0.675	0.385	0.00	1.928	0.420	0.755	0.00	1.547

Method	MCC	FDR	Exact	MSE(θ)	MCC	FDR	Exact	MSE(θ)
ENet-P	0.592	0.522	0.00	1.998	0.356	0.809	0.00	1.595
Dense heterogeneity ($R = 25$)								
MCP-H	0.893	0.150	0.28	0.289	0.794	0.330	0.12	0.220
SCAD-H	0.809	0.289	0.20	0.283	0.695	0.463	0.08	0.225
LASSO-H	0.645	0.503	0.04	0.606	0.428	0.749	0.00	0.345
MCP-S	0.912	0.021	0.52	0.790	0.903	0.132	0.40	0.703
MCP-P	0.841	0.017	0.24	1.080	0.854	0.210	0.24	0.747
SCAD-P	0.860	0.057	0.20	1.115	0.690	0.464	0.00	0.747
LASSO-P	0.728	0.369	0.00	1.461	0.436	0.745	0.00	1.073
ENet-P	0.658	0.473	0.00	1.546	0.370	0.800	0.00	1.139
Heavy censoring ($R = 25$)								
MCP-H	0.869	0.017	0.40	0.339	0.932	0.068	0.56	0.187
SCAD-H	0.875	0.076	0.40	0.322	0.863	0.205	0.24	0.215
LASSO-H	0.613	0.558	0.00	0.682	0.405	0.779	0.00	0.365
MCP-S	0.824	0.026	0.32	0.497	0.916	0.070	0.52	0.296
MCP-P	0.806	0.022	0.12	0.652	0.792	0.321	0.16	0.266
SCAD-P	0.805	0.130	0.12	0.646	0.628	0.554	0.00	0.252
LASSO-P	0.692	0.409	0.04	1.004	0.422	0.770	0.00	0.533
ENet-P	0.614	0.536	0.00	1.111	0.360	0.816	0.00	0.607
Weak signals ($R = 25$)								
MCP-H	0.672	0.160	0.00	0.260	0.646	0.368	0.00	0.197
SCAD-H	0.624	0.274	0.00	0.294	0.586	0.521	0.00	0.178
LASSO-H	0.587	0.434	0.00	0.355	0.466	0.679	0.00	0.225
MCP-S	0.637	0.100	0.04	0.395	0.661	0.291	0.00	0.304
MCP-P	0.460	0.026	0.00	0.607	0.607	0.437	0.00	0.297
SCAD-P	0.538	0.069	0.00	0.619	0.558	0.541	0.00	0.300
LASSO-P	0.527	0.240	0.00	0.601	0.454	0.675	0.00	0.415
ENet-P	0.508	0.296	0.00	0.619	0.425	0.719	0.00	0.432
p = 200 ($R = 25$)								
MCP-H	0.938	0.085	0.44	0.092	0.869	0.225	0.28	0.064
SCAD-H	0.836	0.254	0.20	0.098	0.674	0.517	0.00	0.058
LASSO-H	0.561	0.650	0.00	0.292	0.355	0.844	0.00	0.171
MCP-S	0.940	0.040	0.56	0.163	0.915	0.147	0.48	0.127
MCP-P	0.882	0.029	0.24	0.268	0.823	0.296	0.24	0.144
SCAD-P	0.879	0.099	0.32	0.283	0.644	0.551	0.04	0.148
LASSO-P	0.690	0.472	0.00	0.492	0.409	0.805	0.00	0.323
ENet-P	0.579	0.616	0.00	0.547	0.344	0.853	0.00	0.370
p = 500 ($R = 25$)								
MCP-H	0.902	0.067	0.40	0.055	0.824	0.297	0.08	0.026
SCAD-H	0.785	0.291	0.04	0.053	0.578	0.639	0.00	0.026
LASSO-H	0.600	0.559	0.00	0.162	0.361	0.845	0.00	0.103
MCP-S	0.879	0.044	0.28	0.077	0.842	0.234	0.16	0.057
MCP-P	0.804	0.030	0.12	0.136	0.761	0.374	0.04	0.060
SCAD-P	0.812	0.073	0.04	0.155	0.544	0.646	0.00	0.064
LASSO-P	0.629	0.464	0.00	0.226	0.382	0.836	0.00	0.156
ENet-P	0.550	0.603	0.00	0.255	0.331	0.876	0.00	0.185
Ten studies ($R = 25$)								

Method	MCC	FDR	Exact	MSE(θ)	MCC	FDR	Exact	MSE(θ)
MCP-H	0.966	0.031	0.76	0.127	0.923	0.131	0.44	0.072
SCAD-H	0.937	0.099	0.56	0.122	0.801	0.324	0.12	0.071
LASSO-H	0.602	0.592	0.00	0.416	0.417	0.768	0.00	0.240
MCP-S	0.965	0.016	0.72	0.204	0.944	0.096	0.60	0.142
MCP-P	0.914	0.011	0.40	0.372	0.908	0.154	0.48	0.160
SCAD-P	0.928	0.049	0.48	0.420	0.812	0.302	0.16	0.161
LASSO-P	0.728	0.433	0.00	0.690	0.424	0.769	0.00	0.381
ENet-P	0.632	0.554	0.00	0.774	0.344	0.827	0.00	0.429
Unequal study sizes ($R = 25$)								
MCP-H	0.958	0.048	0.60	0.174	0.908	0.156	0.36	0.105
SCAD-H	0.913	0.131	0.40	0.170	0.807	0.304	0.28	0.107
LASSO-H	0.618	0.562	0.00	0.453	0.391	0.793	0.00	0.253
MCP-S	0.941	0.023	0.52	0.331	0.944	0.098	0.56	0.265
MCP-P	0.868	0.010	0.32	0.551	0.917	0.142	0.52	0.290
SCAD-P	0.880	0.044	0.28	0.583	0.762	0.380	0.12	0.292
LASSO-P	0.738	0.401	0.04	0.847	0.434	0.756	0.00	0.531
ENet-P	0.636	0.537	0.00	0.920	0.365	0.811	0.00	0.589
Small studies, $p > n_k$ ($R = 25$)								
MCP-H	0.764	0.115	0.08	0.309	0.728	0.292	0.04	0.275
SCAD-H	0.715	0.253	0.00	0.358	0.604	0.551	0.00	0.251
LASSO-H	0.552	0.532	0.00	0.589	0.399	0.790	0.00	0.393
MCP-S	0.766	0.049	0.12	0.361	0.757	0.194	0.04	0.270
MCP-P	0.682	0.046	0.00	0.529	0.666	0.368	0.00	0.270
SCAD-P	0.650	0.195	0.00	0.567	0.532	0.627	0.00	0.258
LASSO-P	0.537	0.468	0.00	0.757	0.417	0.777	0.00	0.511
ENet-P	0.489	0.541	0.00	0.839	0.366	0.821	0.00	0.586